

BdPhO Category C Theory Book

Bangladesh Physics Olympiad (BdPhO) Category C

Generated for Physics Olympiad Preparation

Generated for Physics Olympiad Preparation

Chapter 0

Table of Contents

1	Prerequisites for Physics Olympiads	5
1.1	Algebra	5
1.1.1	Polynomials & Equations	5
1.1.2	Sequences and Series	5
1.1.3	Approximations	6
1.2	Trigonometry	6
1.2.1	Angles and Arc Length	6
1.2.2	Trigonometric Identities	6
1.2.3	Properties of Triangles	6
1.3	Calculus	6
1.3.1	Differentiation	7
1.3.2	Maxima and Minima	7
1.3.3	Integration	7
1.4	Statistics and Data Analysis	7
1.4.1	Central Tendency and Dispersion	8
1.4.2	Experimental Error	8
1.4.3	Challenge Problems	8
2	Vectors and Coordinate Systems	8
2.1	Scalars and Vectors	8
2.2	Unit Vectors and Components	9
2.3	Vector Addition	9
2.4	The Dot Product (Scalar Product)	10
2.5	The Cross Product (Vector Product)	10
2.6	Polar Coordinates	10
2.6.1	Challenge Problems	11
2.6.2	Fully Worked Solutions	11
3	Measurement and Dimensional Analysis	11
3.1	Physical Quantities and SI Units	11
3.1.1	Scientific Notation	12
3.2	Dimensions	12
3.3	Dimensional Analysis	12
3.3.1	Checking Equations	12
3.3.2	Deriving Formulas	12
3.4	Dimensions of Physical Constants	13
3.4.1	Challenge Problems	13
4	Mechanics	13
4.1	Newton's Laws and Free-Body Diagrams	14
4.2	Friction and Drag	14
4.3	Center of Mass (CM)	15
4.4	Momentum and Impulse	15
4.4.1	Collisions	15
4.5	Rotational Motion	15
4.6	Equilibrium	16
4.6.1	Challenge Problems	16
4.6.2	Fully Worked Solutions	16
5	Work, Power, and Energy	17
5.1	Work Done by a Force	17
5.2	The Work-Energy Theorem	17

5.3	Conservative and Non-Conservative Forces	17
5.4	Potential Energy	18
5.5	Conservation of Mechanical Energy	18
5.6	Power	18
5.6.1	Challenge Problems	19
5.6.2	Fully Worked Solutions	19
6	Gravitation	20
6.1	Newton's Law of Gravitation	20
6.2	Gravitational Field and Potential	20
6.3	The Shell Theorem	20
6.4	Gravity Inside and Outside Planets	21
6.5	Escape Velocity and Orbits	21
6.6	Kepler's Laws	22
6.6.1	Challenge Problems	22
7	Oscillation	22
7.1	Simple Harmonic Motion (SHM)	22
7.2	Energy in SHM	23
7.3	Common SHM Systems	23
7.4	Damped Oscillation	24
7.5	Forced Oscillation and Resonance	24
7.5.1	Challenge Problems	24
8	Fluids	25
8.1	Density and Pressure	25
8.2	Fluid Statics	25
8.3	Fluid Dynamics	25
8.4	Viscosity and Poiseuille's Law	26
8.5	Surface Tension	26
8.5.1	Challenge Problems	26
9	Waves & Acoustics	27
9.1	Types of Waves	27
9.2	The Wave Equation and Velocity	27
9.3	Superposition and Interference	27
9.4	Stationary (Standing) Waves	27
9.5	Beats	28
9.6	The Doppler Effect	28
9.7	Diffraction	28
9.7.1	Challenge Problems	29
10	Heat & Thermodynamics	29
10.1	Temperature and Thermal Expansion	29
10.2	Heat and Calorimetry	29
10.3	The Ideal Gas Law and Kinetic Theory	29
10.4	The First Law of Thermodynamics	30
10.5	Heat Engines and the Second Law	30
10.6	Entropy	30
10.6.1	Challenge Problems	31
11	Optics	31
11.1	Geometric Optics: Reflection and Refraction	31
11.2	Lenses and Mirrors	32
11.3	Prisms and Dispersion	32
11.4	Physical Optics: Interference	32
11.5	Physical Optics: Diffraction	33
11.5.1	Challenge Problems	33
11.5.2	Fully Worked Solutions	33
12	Electricity	34

12.1	Electrostatics: Force and Field	34
12.2	Gauss's Law	34
12.3	Electric Potential	34
12.4	Capacitance and Dielectrics	35
12.5	Current, Resistance, and DC Circuits	35
12.6	Kirchhoff's Circuit Laws	35
12.6.1	Challenge Problems	36
13	Magnetism	36
13.1	The Magnetic Field and Lorentz Force	36
13.2	Sources of Magnetic Field	36
13.3	Electromagnetic Induction	37
13.4	Inductance	37
13.5	Alternating Current and Transformers	37
13.5.1	Challenge Problems	38
14	Modern Physics & Cosmology	38
14.1	The Photoelectric Effect	38
14.2	Wave-Particle Duality and the Bohr Model	38
14.3	Special Relativity	39
14.4	Cosmology	39
14.4.1	Challenge Problems	40
15	Appendices	40
15.1	Appendix A: Master Formula Sheet	40
15.2	Appendix B: Bibliography and Further Reading	43
15.3	Appendix C: Index	44

Chapter 1

Prerequisites for Physics Olympiads

Before diving into classical mechanics and electromagnetism, a strong mathematical foundation is absolutely essential. Physics is the study of nature, but mathematics is the language in which it is written. In the Bangladesh Physics Olympiad (BdPhO) Category C, you will frequently encounter problems that require algebraic manipulation, calculus for continuous changes, and trigonometry for spatial reasoning.

This chapter concisely reviews the mathematical tools you must master.

1.1 Algebra

Algebraic fluency is non-negotiable. You must be able to manipulate variables quickly and without errors.

1.1.1 Polynomials & Equations

A polynomial is an expression consisting of variables and coefficients, involving only addition, subtraction, multiplication, and non-negative integer exponents.

Key Formula: Quadratic Formula

For a quadratic equation $ax^2 + bx + c = 0$, the solutions are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The discriminant $\Delta = b^2 - 4ac$ determines the nature of the roots. If $\Delta > 0$, there are two distinct real roots. If $\Delta = 0$, there is one repeated real root. If $\Delta < 0$, the roots are complex.

Olympiad Tip: Vieta's Formulas

For $ax^2 + bx + c = 0$ with roots r_1 and r_2 :

- Sum of roots: $r_1 + r_2 = -\frac{b}{a}$
- Product of roots: $r_1 r_2 = \frac{c}{a}$

These are extremely useful for solving problems without explicitly finding the roots.

1.1.2 Sequences and Series

Physical processes often occur in discrete, repeated steps, leading to sequences.

Arithmetic Progression (AP): A sequence where the difference between consecutive terms is constant (d).

- n -th term: $a_n = a + (n - 1)d$
- Sum of n terms: $S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{n}{2}(a + a_n)$

Geometric Progression (GP): A sequence where the ratio of consecutive terms is constant (r).

- n -th term: $a_n = ar^{n-1}$
- Sum of n terms: $S_n = a \frac{1-r^n}{1-r}$ (for $r \neq 1$)
- Sum to infinity ($|r| < 1$): $S_\infty = \frac{a}{1-r}$

Example: Bouncing Ball

A ball is dropped from height h . It bounces back to $e^2 h$ where $e < 1$. Find the total distance traveled before it comes to rest.

The total distance is $D = h + 2(e^2 h) + 2(e^4 h) + \dots$

$$D = h + 2e^2 h(1 + e^2 + e^4 + \dots) = h + 2e^2 h \left(\frac{1}{1-e^2} \right) = h \left(\frac{1+e^2}{1-e^2} \right)$$

1.1.3 Approximations

In physics, we often deal with quantities that are very small. The first-order Taylor approximation (also known as the binomial approximation for powers) is a powerful tool.

Key Formula: Binomial Approximation

For $|x| \ll 1$ and any real number n :

$$(1 + x)^n \approx 1 + nx$$

Olympiad Tip: Common Approximations

For $x \ll 1$:

- $\sin x \approx x$
- $\cos x \approx 1 - \frac{x^2}{2}$
- $\tan x \approx x$
- $e^x \approx 1 + x$
- $\ln(1 + x) \approx x$

1.2 Trigonometry

Trigonometry is the study of triangles and relationships between their sides and angles. It is essential for vector decomposition and analyzing oscillatory motion.

1.2.1 Angles and Arc Length

An angle θ in radians is defined as the ratio of the arc length s to the radius r of a circle.

$$\theta = \frac{s}{r} \Rightarrow s = r\theta$$

1.2.2 Trigonometric Identities

You must memorize the fundamental identities:

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\sin(2\theta) = 2 \sin \theta \cos \theta$
- $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$

1.2.3 Properties of Triangles

For any triangle with sides a, b, c opposite to angles A, B, C :

Key Formula: Sine and Cosine Rules

Sine Rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine Rule:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

1.3 Calculus

Calculus is the mathematics of change. Mechanics cannot be deeply understood without it, as velocity is the derivative of position, and acceleration is the derivative of velocity.

1.3.1 Differentiation

Differentiation measures the instantaneous rate of change of a function. Geometrically, it is the slope of the tangent line to the curve $y = f(x)$.

Definition: Derivative

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Standard Derivatives:

- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(e^{ax}) = ae^{ax}$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Rules of Differentiation:

- Product Rule: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$
- Quotient Rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$
- Chain Rule: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$

Common Misconception: Constant Acceleration

Many students assume acceleration is always constant and use formulas like $v = u + at$. If acceleration depends on time $a(t)$ or position $a(x)$, you **must** use calculus: $v(t) = v_0 + \int a(t)dt$.

1.3.2 Maxima and Minima

To find the maximum or minimum of a function $f(x)$, we set its first derivative to zero: $f'(x) = 0$. To determine if it is a maximum or minimum, we check the second derivative $f''(x)$:

- If $f''(x) > 0$, it is a local minimum.
- If $f''(x) < 0$, it is a local maximum.

1.3.3 Integration

Integration is the reverse process of differentiation. It represents the accumulation of a quantity, geometrically equivalent to the area under the curve $y = f(x)$.

Key Formula: Fundamental Theorem of Calculus

$$\int_a^b f(x)dx = F(b) - F(a)$$

where $F(x)$ is the antiderivative of $f(x)$, meaning $F'(x) = f(x)$.

Standard Integrals:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ (for $n \neq -1$)
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int e^{ax} dx = \frac{1}{a}e^{ax} + C$

1.4 Statistics and Data Analysis

Physics is an experimental science. Experimental data always contains uncertainties, and statistics help us make sense of this data.

1.4.1 Central Tendency and Dispersion

- **Mean (μ):** The average value of a dataset. $\mu = \frac{1}{N} \sum x_i$
- **Median:** The middle value when the data is sorted. Less sensitive to outliers than the mean.
- **Standard Deviation (σ):** Measures the spread of data around the mean.

$$\sigma = \sqrt{\frac{1}{N} \sum (x_i - \mu)^2}$$

1.4.2 Experimental Error

Every measurement has an uncertainty.

- **Absolute Error (Δx):** The magnitude of uncertainty in a measurement.
- **Relative (Fractional) Error:** $\frac{\Delta x}{x}$
- **Percentage Error:** $\frac{\Delta x}{x} \times 100\%$

Error Propagation: If $z = x + y$ or $z = x - y$, then $\Delta z = \delta x + \delta y$. If $z = xy$ or $z = \frac{x}{y}$, then $\frac{\Delta z}{z} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$. If $z = x^n$, then $\frac{\Delta z}{z} = n \frac{\Delta x}{x}$.

Chapter Summary

- **Algebra** provides the tools for exact symbolic manipulation and powerful approximations (e.g., $(1+x)^n \approx 1+nx$).
- **Trigonometry** is crucial for 2D and 3D geometric analysis. Know the sine and cosine rules perfectly.
- **Calculus** (differentiation and integration) is the foundation of kinematics and continuous physics.
- **Statistics** enables the rigorous analysis of experimental data and uncertainties.

1.4.3 Challenge Problems

1. **Beginner:** Approximate the value of $\sqrt{1.02}$ using the binomial approximation.
2. **Intermediate:** Find the maximum area of a rectangle that can be inscribed in a semicircle of radius R .
3. **Advanced (Olympiad):** A function $f(x)$ represents the potential energy of a particle, $f(x) = Ax^4 - Bx^2$. Find the equilibrium positions (where force is zero) and determine if they are stable (minimum energy) or unstable (maximum energy).

Chapter 2

Vectors and Coordinate Systems

In physics, quantities are generally classified into two categories: scalars and vectors. A deep geometric and algebraic understanding of vectors is the most fundamental prerequisite for mechanics and electromagnetism.

2.1 Scalars and Vectors

Definition: Scalars and Vectors

- **Scalar:** A physical quantity that is completely described by a magnitude (and a unit). Examples: Mass, temperature, time, speed, energy.
- **Vector:** A physical quantity that has both a magnitude and a direction, and obeys the rules of vector addition. Examples: Displacement, velocity, acceleration, force, momentum.

A vector is typically denoted in boldface (e.g., $\mathbf{A}$) or with an arrow over it ($\vec{A}$). Its magnitude is denoted as $|\mathbf{A}|$ or simply A .

Geometrically, a vector is represented by an arrow. The length of the arrow is proportional to the vector's magnitude, and it points in the direction of the vector.

Common Misconception: Direction is not enough

Not all quantities with direction are vectors. Finite rotations have magnitude and direction but do not commute under addition (i.e., $\theta_1 + \theta_2 \neq \theta_2 + \theta_1$), so they are not vectors!

2.2 Unit Vectors and Components

A unit vector is a vector with a magnitude of exactly 1. It is used solely to specify a direction. The unit vector in the direction of $\mathbf{A}$ is denoted with a “tilde” ($\tilde{A}$):

$$\tilde{A} = \frac{\mathbf{A}}{|\mathbf{A}|} \Rightarrow \mathbf{A} = |\mathbf{A}| \tilde{A}$$

In a 3D Cartesian coordinate system, the standard basis unit vectors along the x , y , and z axes are $\tilde{i}$, $\tilde{j}$, and $\tilde{k}$ respectively.

Any vector $\mathbf{A}$ can be decomposed into scalar components along these axes:

$$\mathbf{A} = A_x \tilde{i} + A_y \tilde{j} + A_z \tilde{k}$$

The magnitude of $\mathbf{A}$ is given by the 3D Pythagorean theorem:

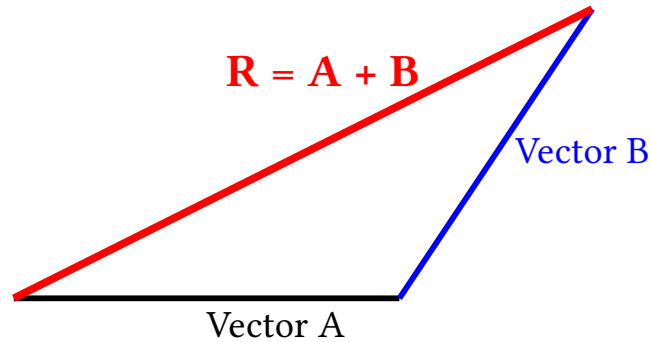
$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

2.3 Vector Addition

Vectors add geometrically via the “tail-to-tip” method or the parallelogram rule.

Algebraically, if $\mathbf{A} = A_x \tilde{i} + A_y \tilde{j} + A_z \tilde{k}$ and $\mathbf{B} = B_x \tilde{i} + B_y \tilde{j} + B_z \tilde{k}$, their sum is:

$$\mathbf{A} + \mathbf{B} = (A_x + B_x) \tilde{i} + (A_y + B_y) \tilde{j} + (A_z + B_z) \tilde{k}$$

**Key Formula: Law of Cosines for Vectors**

The magnitude of the sum $\mathbf{R} = \mathbf{A} + \mathbf{B}$ when the angle between their tails is θ is:

$$|\mathbf{R}| = \sqrt{|\mathbf{A}|^2 + |\mathbf{B}|^2 + 2 |\mathbf{A}| |\mathbf{B}| \cos \theta}$$

Notice the $+2$ instead of -2 from standard geometry, because the angle θ here is the exterior angle of the vector addition triangle.

2.4 The Dot Product (Scalar Product)

The dot product multiplies two vectors to yield a scalar. It measures how much of one vector goes in the direction of the other.

Definition: Dot Product

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta$$

where θ is the angle between the two vectors.

Algebraically, in component form:

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

Physical Intuition: The dot product is used to compute work (Work = $\mathbf{F} \cdot \mathbf{d}$) and electric/magnetic flux. If $\mathbf{A} \cdot \mathbf{B} = 0$, the vectors are perfectly perpendicular (orthogonal).

2.5 The Cross Product (Vector Product)

The cross product multiplies two vectors to yield a third vector that is perpendicular to both.

Definition: Cross Product

$$\mathbf{C} = \mathbf{A} \times \mathbf{B}$$

The magnitude is $|\mathbf{C}| = |\mathbf{A}| |\mathbf{B}| \sin \theta$. The direction is given by the **right-hand rule**.

Right-Hand Rule: Point the fingers of your right hand in the direction of $\mathbf{A}$, curl them towards $\mathbf{B}$. Your thumb points in the direction of $\mathbf{A} \times \mathbf{B}$.

Algebraically, it is evaluated as a determinant:

$$\mathbf{A} \times \mathbf{B} = \det \begin{pmatrix} \tilde{i} & \tilde{j} & \tilde{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{pmatrix} = (A_y B_z - A_z B_y) \tilde{i} - (A_x B_z - A_z B_x) \tilde{j} + (A_x B_y - A_y B_x) \tilde{k}$$

Properties:

- Anti-commutative: $\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$
- Parallel test: If $\mathbf{A} \times \mathbf{B} = 0$, the vectors are parallel or anti-parallel.

Physical Intuition: The cross product is essential for rotational quantities. Torque is $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$, and angular momentum is $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. It also describes the magnetic force on a moving charge: $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$.

2.6 Polar Coordinates

While Cartesian coordinates (x, y) specify a point by its projection onto straight axes, polar coordinates (r, θ) specify a point in 2D space by its distance r from the origin and the angle θ measured counterclockwise from the positive x-axis.

Key Formula: Coordinate Conversion

From Polar to Cartesian:

$$x = r \cos \theta, \quad y = r \sin \theta$$

From Cartesian to Polar:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right)$$

Unit Vectors in Polar Coordinates: Instead of static $\tilde{i}$ and $\tilde{j}$, polar coordinates use unit vectors $\tilde{r}$ (pointing outward) and $\tilde{\theta}$ (pointing counterclockwise). These unit vectors change direction depending on the position of the particle!

$$\tilde{r} = \cos \theta \tilde{i} + \sin \theta \tilde{j}$$

$$\tilde{\theta} = -\sin \theta \tilde{i} + \cos \theta \tilde{j}$$

When differentiating position $\mathbf{r} = r\tilde{r}$ to find velocity, you must also differentiate the unit vector $\tilde{r}$ because it changes with time if θ changes. This is the origin of centripetal and Coriolis accelerations.

Olympiad Tip: When to use Polar Coordinates

Any problem involving central forces (like gravity or electrostatic force) or circular motion is almost always solved more easily in polar coordinates rather than Cartesian.

Chapter Summary

- **Vectors** require both magnitude and direction, and obey specific addition rules.
- Use **components** for rigid algebraic manipulation, but remember the geometry.
- **Dot product** ($\cos \theta$) yields a scalar and tests for perpendicularity.
- **Cross product** ($\sin \theta$) yields a perpendicular vector via the right-hand rule and tests for parallel alignment.
- **Polar coordinates** use position-dependent unit vectors $\tilde{r}$ and $\tilde{\theta}$, ideal for rotational physics.

2.6.1 Challenge Problems

1. **Beginner:** Given $\mathbf{A} = 3\tilde{i} + 4\tilde{j}$ and $\mathbf{B} = 5\tilde{i} - 12\tilde{j}$, find the angle between them.
2. **Intermediate:** Prove that $|\mathbf{A} \times \mathbf{B}|^2 + (\mathbf{A} \cdot \mathbf{B})^2 = |\mathbf{A}|^2 |\mathbf{B}|^2$.
3. **Advanced (Olympiad):** Three vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$ sum to zero. Show that $\mathbf{A} \times \mathbf{B} = \mathbf{B} \times \mathbf{C} = \mathbf{C} \times \mathbf{A}$. What is the geometric interpretation of this result? (Hint: Think about the area of a triangle).

2.6.2 Fully Worked Solutions

Solution to 1: First, find the dot product: $\mathbf{A} \cdot \mathbf{B} = (3)(5) + (4)(-12) = 15 - 48 = -33$. Next, find magnitudes: $|\mathbf{A}| = \sqrt{3^2 + 4^2} = 5$ and $|\mathbf{B}| = \sqrt{5^2 + (-12)^2} = 13$. Using the dot product formula: $\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}| |\mathbf{B}|} = -\frac{33}{65}$. Thus, $\theta = \arccos\left(-\frac{33}{65}\right) \approx 120.5^\circ$.

Solution to 2: By definition: $|\mathbf{A} \times \mathbf{B}|^2 = (|\mathbf{A}| |\mathbf{B}| \sin \theta)^2$ and $(\mathbf{A} \cdot \mathbf{B})^2 = (|\mathbf{A}| |\mathbf{B}| \cos \theta)^2$. Summing them: $|\mathbf{A}|^2 |\mathbf{B}|^2 (\sin^2 \theta + \cos^2 \theta)$. Since $\sin^2 \theta + \cos^2 \theta = 1$, the expression simplifies to $|\mathbf{A}|^2 |\mathbf{B}|^2$. This is known as Lagrange's identity.

Solution to 3: Given $\mathbf{A} + \mathbf{B} + \mathbf{C} = 0$, we can write $\mathbf{C} = -(\mathbf{A} + \mathbf{B})$. Substitute this into $\mathbf{B} \times \mathbf{C}$: $\mathbf{B} \times \mathbf{C} = \mathbf{B} \times (-\mathbf{A} - \mathbf{B}) = -(\mathbf{B} \times \mathbf{A}) - (\mathbf{B} \times \mathbf{B})$. Since the cross product of a vector with itself is zero ($\mathbf{B} \times \mathbf{B} = 0$), and it is anti-commutative ($-(\mathbf{B} \times \mathbf{A}) = \mathbf{A} \times \mathbf{B}$), we get: $\mathbf{B} \times \mathbf{C} = \mathbf{A} \times \mathbf{B}$. By symmetric reasoning, $\mathbf{C} \times \mathbf{A} = \mathbf{A} \times \mathbf{B}$.

Geometric Interpretation: The magnitude of the cross product of two adjacent sides of a triangle is twice the area of the triangle. Since $\mathbf{A}, \mathbf{B}$, and $\mathbf{C}$ form a closed triangle, taking the cross product of any two sides yields the same vector (perpendicular to the plane, magnitude proportional to the area).

Chapter 3

Measurement and Dimensional Analysis

Physics is fundamentally an experimental science, meaning all theories must be tested by measurement. How we measure, quantify, and analyze the units of physical quantities forms the basis of all physical equations.

3.1 Physical Quantities and SI Units

A physical quantity is any property of a physical system that can be quantified by measurement. The International System of Units (SI) is the universally accepted standard.

Definition: Fundamental and Derived Quantities

- **Fundamental Quantities:** Seven independent quantities from which all others are derived.
 1. Mass (Kilogram, kg)
 2. Length (Meter, m)
 3. Time (Second, s)
 4. Electric Current (Ampere, A)
 5. Temperature (Kelvin, K)
 6. Amount of Substance (Mole, mol)
 7. Luminous Intensity (Candela, cd)
- **Derived Quantities:** Formed by multiplying or dividing fundamental quantities. Examples include Velocity (m/s), Force ($N = kg\ m/s^2$), and Energy ($J = kg\ m^2/s^2$).

3.1.1 Scientific Notation

Physical quantities span enormous scales. The mass of an electron is roughly 9.11×10^{-31} kg, while the mass of the Sun is 1.989×10^{30} kg. Scientific notation, $a \times 10^b$ where $1 \leq a < 10$, is essential for expressing these values compactly and identifying the number of significant figures.

3.2 Dimensions

The “dimension” of a physical quantity denotes its physical nature, independent of the specific unit used (e.g., meters, inches, or light-years all have the dimension of length).

We represent the dimensions of the fundamental quantities using bracketed capital letters:

- Mass: $[M]$
- Length: $[L]$
- Time: $[T]$
- Current: $[I]$
- Temperature: $[\theta]$

The dimension of any derived quantity is found by substituting these into its defining formula. For example, velocity is length divided by time, so its dimension is $[v] = [L][T]^{-1}$. Acceleration is $[a] = [L][T]^{-2}$. Force is mass times acceleration, so $[F] = [M][L][T]^{-2}$.

3.3 Dimensional Analysis

Dimensional analysis is one of the most powerful tools in a physicist’s arsenal. It relies on the **Principle of Dimensional Homogeneity**, which states that every valid physical equation must have the same dimensions on both sides. You cannot add mass to length!

Olympiad Tip: The Power of Dimensional Analysis

If you forget a formula during the Olympiad, or if you need to guess the relationship between variables in a system you don’t fully understand, dimensional analysis can often provide the answer up to a dimensionless constant.

3.3.1 Checking Equations

If you derive a formula for the period of a pendulum as $T = 2\pi\sqrt{\frac{g}{l}}$, you can check it using dimensions:

- Left side: $[T] = [T]$
- Right side: $[\sqrt{\frac{g}{l}}] = \sqrt{\frac{[L][T]^{-2}}{[L]}} = \sqrt{[T]^{-2}} = [T]^{-1}$

Since $[T] \neq [T]^{-1}$, the formula is incorrect! (The correct formula is $T = 2\pi\sqrt{\frac{l}{g}}$).

3.3.2 Deriving Formulas

Suppose we want to find the centripetal force F on a particle of mass m moving with speed v in a circle of radius r . We assume:

$$F = km^x v^y r^z$$

where k is a dimensionless constant. Taking dimensions on both sides:

$$[M][L][T]^{-2} = [M]^x([L][T]^{-1})^y[L]^z$$

$$[M]^1[L]^1[T]^{-2} = [M]^x[L]^{y+z}[T]^{-y}$$

Equating the exponents of corresponding dimensions:

- For M : $x = 1$
- For T : $-y = -2 \Rightarrow y = 2$
- For L : $y + z = 1 \Rightarrow 2 + z = 1 \Rightarrow z = -1$

Thus, $F = k \frac{mv^2}{r}$. Dimensional analysis correctly predicts the functional form!

3.4 Dimensions of Physical Constants

Physical constants are not always dimensionless numbers; they have dimensions necessary to balance equations.

Example: Universal Gravitational Constant

Newton's Law of Gravitation is $F = G \frac{m_1 m_2}{r^2}$. What are the dimensions of G ?

Rearranging for G : $G = \frac{Fr^2}{m_1 m_2}$

Substitute dimensions:

$$[G] = \frac{[M][L][T]^{-2}[L]^2}{[M]^2} = [M]^{-1}[L]^3[T]^{-2}$$

Other important constants:

- Planck's constant (h): $E = hf \Rightarrow [h] = [M][L]^2[T]^{-1}$ (same as angular momentum)
- Permittivity of free space (ϵ_0): $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \Rightarrow [\epsilon_0] = [M]^{-1}[L]^{-3}[T]^4[I]^2$

Chapter Summary

- **Fundamental quantities** form the basis of all physical measurement (M, L, T, I, θ , etc.).
- **Dimensional homogeneity** dictates that you can only equate or add terms with identical dimensions.
- **Dimensional analysis** can be used to check derived equations, uncover dependencies, and determine the units of unknown physical constants.

3.4.1 Challenge Problems

1. **Beginner:** Find the dimensions of kinetic energy ($\frac{1}{2}mv^2$) and potential energy (mgh) and verify they are the same.
2. **Intermediate:** The velocity v of waves on a string depends on the tension F in the string and its mass per unit length μ . Use dimensional analysis to find the formula for v (ignore dimensionless constants).
3. **Advanced (Olympiad):** In fluid mechanics, the drag force F_D on a sphere moving through a fluid depends on its radius r , velocity v , and the fluid's viscosity η . Given that $[\eta] = [M][L]^{-1}[T]^{-1}$, find the expression for the drag force. This result is known as Stokes' Law.

Chapter 4 Mechanics

Mechanics is the heart of classical physics. It describes how and why things move. In the Olympiad context, mechanics problems often combine multiple concepts—such as collisions followed by rotational motion—requiring a deep, unified understanding of forces, momentum, and torques.

4.1 Newton's Laws and Free-Body Diagrams

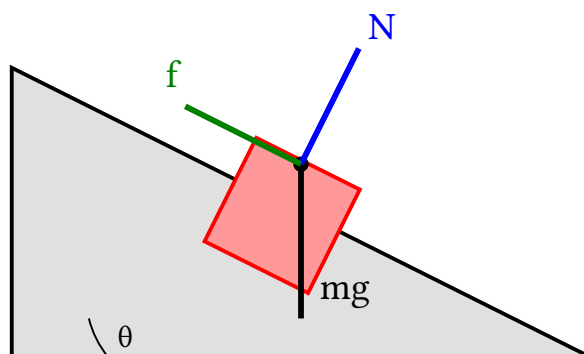
Definition: Inertia and Force

Inertia is the inherent property of matter that resists changes in its state of motion. Mass is the quantitative measure of inertia. A **Force** is a push or pull that can cause an object with mass to accelerate.

Newton's three laws govern classical mechanics:

1. **First Law:** An object remains at rest or in uniform motion unless acted upon by a net external force ($\sum \mathbf{F} = 0 \Rightarrow v = \text{constant}$).
2. **Second Law:** The rate of change of momentum is proportional to the net force applied: $\sum \mathbf{F} = \frac{d\mathbf{p}}{dt}$. If mass is constant, this simplifies to $\sum \mathbf{F} = m\mathbf{a}$.
3. **Third Law:** For every action, there is an equal and opposite reaction ($\mathbf{F}_{12} = -\mathbf{F}_{21}$).

Free-Body Diagrams (FBDs): The single most important skill in mechanics is drawing a correct FBD. An FBD isolates an object and shows **only** the external forces acting **on** it.



Olympiad Tip: Drawing FBDs

1. Isolate the body. Do not draw forces the body exerts on other things.
2. Draw gravity (mg) straight down.
3. Draw normal forces perpendicular to surfaces.
4. Draw tension along strings, pulling away from the object.
5. Draw friction parallel to surfaces, opposing relative slipping.

4.2 Friction and Drag

Dry Friction: Friction arises from microscopic surface interactions.

- **Static Friction (f_s):** Prevents relative motion. It adjusts to exactly balance applied forces up to a maximum limit: $f_s \leq \mu_s N$, where μ_s is the coefficient of static friction and N is the normal force.
- **Kinetic Friction (f_k):** Opposes ongoing relative motion. It is approximately constant: $f_k = \mu_k N$. Usually, $\mu_k < \mu_s$.

Drag Force: When an object moves through a fluid (like air or water), it experiences drag. For objects moving fast (like a thrown ball), the drag is proportional to velocity squared:

$$F_D = \frac{1}{2} \rho v^2 C_D A$$

where ρ is fluid density, v is velocity, C_D is the drag coefficient, and A is cross-sectional area. When drag equals gravity, acceleration becomes zero, and the object reaches **terminal velocity**.

4.3 Center of Mass (CM)

The center of mass is the unique point where the entire mass of a system can be considered to be concentrated for the purpose of analyzing translational motion.

Key Formula: Center of Mass

For a system of discrete particles:

$$\mathbf{r}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i} = \frac{1}{M} \sum m_i \mathbf{r}_i$$

For a continuous body:

$$\mathbf{r}_{\text{cm}} = \frac{1}{M} \int \mathbf{r} dm$$

An external force acting on a system accelerates its CM exactly as if all mass were concentrated there: $\sum \mathbf{F}_{\text{ext}} = M \mathbf{a}_{\text{cm}}$. Internal forces (like an explosion) do not affect the motion of the CM.

4.4 Momentum and Impulse

Linear momentum is $\mathbf{p} = m\mathbf{v}$. **Impulse** ($\mathbf{J}$) is the change in momentum caused by a force acting over a time interval Δt :

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F} dt = \Delta \mathbf{p}$$

4.4.1 Collisions

During any collision where external impulsive forces are zero, the total momentum of the system is conserved: $\mathbf{P}_{\text{initial}} = \mathbf{P}_{\text{final}}$.

- **Elastic Collisions:** Both momentum and kinetic energy are conserved. The relative velocity of approach equals the relative velocity of separation.
- **Inelastic Collisions:** Momentum is conserved, but kinetic energy is lost (usually to heat or deformation).
- **Perfectly Inelastic Collisions:** The bodies stick together after impact. Kinetic energy loss is maximum.

Common Misconception: Momentum vs Energy Conservation

Momentum is conserved in **all** isolated collisions. Mechanical energy is **only** conserved in perfectly elastic collisions. Do not use energy conservation across an inelastic collision!

4.5 Rotational Motion

Rotational kinematics mirror linear kinematics.

- Position $x \rightarrow$ Angle θ
- Velocity $v \rightarrow$ Angular velocity $\omega = \frac{d\theta}{dt}$
- Acceleration $a \rightarrow$ Angular acceleration $\alpha = \frac{d\omega}{dt}$

Torque (τ): The rotational equivalent of force. It measures the tendency of a force to cause rotation about an axis.

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$$

Magnitude: $\tau = rF \sin \varphi = Fd_{\perp}$ (where $d_{\perp}$ is the lever arm).

Moment of Inertia (I): The rotational equivalent of mass. It depends not just on the mass, but how it is distributed relative to the axis of rotation.

$$I = \sum m_i r_i^2 \quad \text{or} \quad I = \int r^2 dm$$

Key Formula: Parallel and Perpendicular Axis Theorems

Parallel Axis Theorem: Connects I through the CM to any parallel axis at distance d :

$$I = I_{\text{cm}} + Md^2$$

Perpendicular Axis Theorem (for flat 2D laminae only):

$$I_z = I_x + I_y$$

Angular Momentum (L): The rotational equivalent of linear momentum. For a point particle: $L = r \times p$ For a rigid body rotating about a symmetry axis: $L = I\omega$

Newton's Second Law for rotation states that the net external torque equals the rate of change of angular momentum: $\sum \tau = \frac{dL}{dt} = I\alpha$.

4.6 Equilibrium

For a rigid body to be in static equilibrium, it must not accelerate linearly or rotationally.

1. Translational equilibrium: $\sum \mathbf{F} = 0$
2. Rotational equilibrium: $\sum \tau = 0$ (about **any** chosen pivot point).

Olympiad Tip: Choosing a Pivot Point

When summing torques, you can choose any point as your pivot. A smart choice is a point where unknown forces act, as their lever arm becomes zero and they drop out of the torque equation!

Chapter Summary

- FBDs are essential for applying $\sum \mathbf{F} = m\mathbf{a}$.
- **Friction** opposes slipping; **Drag** opposes motion through a fluid.
- The **Center of Mass** moves as if all external forces act directly on it.
- **Momentum** is conserved if net external force is zero.
- **Angular momentum** is conserved if net external torque is zero.
- Use $\sum \tau = 0$ about strategic pivots to solve equilibrium problems.

4.6.1 Challenge Problems

1. **Beginner:** A wheel of radius R and mass M accelerates from rest to angular velocity ω in time t . Find the constant torque applied.
2. **Intermediate:** A bullet of mass m is fired with velocity v into a wooden block of mass M hanging from a string of length L . It embeds itself. Find the maximum angle the string swings.
3. **Advanced (Olympiad):** A solid cylinder of mass M and radius R is placed on a rough inclined plane of angle θ . Find the minimum coefficient of static friction μ_s required for the cylinder to roll without slipping. (Note: $I_{\text{cm}} = \frac{1}{2}MR^2$).

4.6.2 Fully Worked Solutions

Solution to 1: Angular acceleration is $\alpha = \frac{\omega-0}{t} = \frac{\omega}{t}$. Assuming the wheel is a solid disk, its moment of inertia is $I = \frac{1}{2}MR^2$. Using Newton's Second Law for rotation: $\tau = I\alpha = \left(\frac{1}{2}MR^2\right)\left(\frac{\omega}{t}\right)$.

Solution to 2: This is a classic ballistic pendulum problem. It happens in two distinct stages. **Stage 1: Inelastic Collision.** The bullet hits the block. Momentum is conserved, but mechanical energy is **not**. $mv = (m+M)V \Rightarrow V = \frac{mv}{m+M}$ **Stage 2: Swinging up.** The combined mass swings upward. Now, gravity and tension are the only forces (tension does no work). Mechanical energy is conserved. $\frac{1}{2}(m+M)V^2 = (m+M)gh$ Substitute V : $\frac{1}{2}\left(\frac{mv}{m+M}\right)^2 = gh \Rightarrow h = \frac{m^2v^2}{2g(m+M)^2}$ From geometry, the height h relates to the angle φ by $h = L - L \cos \varphi = L(1 - \cos \varphi)$. $L(1 - \cos \varphi) = \frac{m^2v^2}{2g(m+M)^2} \Rightarrow \varphi = \arccos\left(1 - \frac{m^2v^2}{2gL(m+M)^2}\right)$

Solution to 3: Let's draw a Free-Body Diagram. The forces on the cylinder are gravity (Mg downwards), normal force (N perpendicular to the plane), and static friction (f_s acting up the incline to cause torque).

1. Translational equation down the incline: $Mg \sin \theta - f_s = Ma$
2. Translational equation perpendicular to incline: $N - Mg \cos \theta = 0 \Rightarrow N = Mg \cos \theta$

3. Rotational equation about the CM: The only force causing torque is friction. $\tau = f_s R = I\alpha$.

Since it rolls without slipping, $a = \alpha R$. Substitute $\alpha = \frac{a}{R}$ and $I = \frac{1}{2}MR^2$: $f_s R = (\frac{1}{2}MR^2)(\frac{a}{R}) \Rightarrow f_s = \frac{1}{2}Ma$
 Substitute this f_s into the first equation: $Mg \sin \theta - \frac{1}{2}Ma = Ma \Rightarrow Mg \sin \theta = \frac{3}{2}Ma \Rightarrow a = \frac{2}{3}g \sin \theta$ Now find the required friction: $f_s = \frac{1}{2}M(\frac{2}{3}g \sin \theta) = \frac{1}{3}Mg \sin \theta$. To prevent slipping, we must have $f_s \leq \mu_s N$. $\frac{1}{3}Mg \sin \theta \leq \mu_s (Mg \cos \theta)$ $\mu_s \geq \frac{1}{3} \tan \theta$. The minimum coefficient is $\frac{1}{3} \tan \theta$.

Chapter 5

Work, Power, and Energy

Newtonian mechanics allows us to determine the exact trajectory of a particle if we know all the forces acting on it at all times. However, in many complex systems, tracking forces continuously is impossible. The concepts of Work and Energy provide a powerful alternative framework based on conservation laws.

5.1 Work Done by a Force

In physics, work is only done when a force causes a displacement, and only the component of the force parallel to the displacement contributes to the work.

Definition: Work

For a constant force $\mathbf{F}$ causing a displacement $\Delta \mathbf{r}$, work W is the dot product:

$$W = \mathbf{F} \cdot \Delta \mathbf{r} = F \Delta r \cos \theta$$

where θ is the angle between the force and the displacement.

Work by a Variable Force: If the force changes with position (e.g., a stretching spring where $F = -kx$), we must integrate the force over the path from the initial position r_1 to the final position r_2 :

$$W = \int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r}$$

Graphically, in one dimension, work is the area under the Force vs. Position (F - x) curve.

5.2 The Work-Energy Theorem

Kinetic Energy (K) is the energy of motion: $K = \frac{1}{2}mv^2$.

The central theorem linking forces to energy is the Work-Energy Theorem. It states that the net work done on an object by **all** forces (both conservative and non-conservative) equals its change in kinetic energy.

Key Formula: Work-Energy Theorem

$$W_{\text{net}} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Olympiad Tip: When to use Work-Energy

Use the Work-Energy theorem when a problem involves speed and distance, but doesn't ask about time. It bypasses the need to find acceleration!

5.3 Conservative and Non-Conservative Forces

Forces in nature can be divided into two categories based on the work they do.

Conservative Forces: A force is conservative if the work it does on an object moving between two points is independent of the path taken. Equivalently, the work done in any closed loop is zero ($\oint \mathbf{F} \cdot d\mathbf{r} = 0$). Examples: Gravity, Electrostatic force, Spring force.

Non-Conservative Forces: A force is non-conservative if the work done depends on the path taken. These forces typically dissipate mechanical energy into thermal energy (heat). Examples: Friction, Air resistance (drag), Tension, Normal force (though normal force usually does zero work since it's perpendicular to motion).

5.4 Potential Energy

Because the work done by a conservative force is path-independent, we can associate a **Potential Energy** (U) with the position of the object in the force field.

We define the change in potential energy ΔU as the **negative** of the work done by the conservative force:

$$\Delta U = -W_c = - \int_{r_1}^{r_2} \mathbf{F}_c \cdot d\mathbf{r}$$

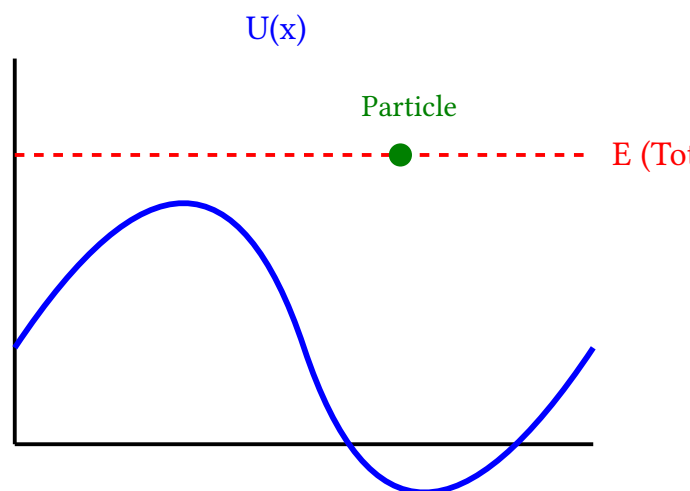
Conversely, if we know the potential energy function $U(\mathbf{r})$, we can find the conservative force by taking the negative gradient (derivative) of U :

$$\mathbf{F}_c = -\nabla U$$

In one dimension: $F_x = -\frac{dU}{dx}$. **Intuition:** Objects naturally tend to accelerate towards positions of lower potential energy (downhill).

Common Potential Energies:

- Uniform Gravity (near Earth's surface): $U_g = mgh$
- Ideal Spring: $U_s = \frac{1}{2}kx^2$



5.5 Conservation of Mechanical Energy

The total mechanical energy E of a system is the sum of its kinetic and potential energies: $E = K + U$.

If **only** conservative forces do work on a system, the total mechanical energy is conserved.

$$E_i = E_f \Rightarrow K_i + U_i = K_f + U_f$$

If non-conservative forces (like friction) also do work, the mechanical energy changes by the amount of work done by those forces:

$$W_{nc} = \Delta E = (K_f + U_f) - (K_i + U_i)$$

5.6 Power

Power is the rate at which work is done or energy is transferred.

Key Formula: Power

$$\begin{aligned}\text{Average Power: } P_{\text{avg}} &= \frac{\Delta W}{\Delta t} \\ \text{Instantaneous Power: } P &= \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v}\end{aligned}$$

The unit of power is the Watt ($W = J/s$).

Chapter Summary

- **Work** is the dot product of force and displacement. For variable forces, it is the integral $\int \mathbf{F} \cdot d\mathbf{r}$.
- The **Work-Energy Theorem** relates net work to the change in Kinetic Energy.
- **Conservative forces** (gravity, springs) have an associated **Potential Energy** ($\Delta U = -W_c$).
- **Mechanical energy** ($K + U$) is conserved if no non-conservative forces do work.
- **Power** is the dot product of force and velocity ($\mathbf{F} \cdot \mathbf{v}$).

5.6.1 Challenge Problems

1. **Beginner:** A block of mass m slides down a rough inclined plane of height h and angle θ . If the coefficient of kinetic friction is μ_k , find its speed at the bottom using energy methods.
2. **Intermediate:** A particle moves in 1D under a potential energy $U(x) = ax^2 - bx^3$. Find the equilibrium positions and determine whether they are stable or unstable.
3. **Advanced (Olympiad):** A chain of length L and mass M is held vertically such that its lower end just touches a scale. The chain is released and falls freely. Find the reading on the scale as a function of the length of the chain x that has hit the scale. (Hint: The scale reading includes both the weight of the resting chain and the rate of change of momentum of the falling links).

5.6.2 Fully Worked Solutions

Solution to 1: Using the Work-Energy theorem with non-conservative forces: $W_{\text{nc}} = \Delta E = E_f - E_i$. The initial energy at the top is pure potential: $E_i = mgh$. The final energy at the bottom is pure kinetic: $E_f = \frac{1}{2}mv^2$. The non-conservative force is friction $f_k = \mu_k N = \mu_k mg \cos \theta$. The distance along the incline is $d = \frac{h}{\sin \theta}$. Work done by friction is negative: $W_{\text{nc}} = -f_k d = -(\mu_k mg \cos \theta) \left(\frac{h}{\sin \theta} \right) = -\mu_k mgh \cot \theta$. $-\mu_k mgh \cot \theta = \frac{1}{2}mv^2 - mgh$. $\frac{1}{2}mv^2 = mgh(1 - \mu_k \cot \theta) \Rightarrow v = \sqrt{2gh(1 - \mu_k \cot \theta)}$.

Solution to 2: Equilibrium points occur where the net force is zero. Force is the negative gradient of potential energy: $F(x) = -\frac{dU}{dx} = -(2ax - 3bx^2)$. Set $F(x) = 0$: $2ax - 3bx^2 = x(2a - 3bx) = 0$. The equilibrium points are $x = 0$ and $x = \frac{2a}{3b}$. To check stability, we look at the second derivative $\frac{d^2U}{dx^2} = 2a - 6bx$.

- At $x = 0$: $\frac{d^2U}{dx^2} = 2a$. Assuming $a > 0$, the second derivative is positive. This means $U(x)$ is concave up (a local minimum), so $x = 0$ is a **stable** equilibrium.
- At $x = \frac{2a}{3b}$: $\frac{d^2U}{dx^2} = 2a - 6b\left(\frac{2a}{3b}\right) = 2a - 4a = -2a$. This is negative (a local maximum), so $x = \frac{2a}{3b}$ is an **unstable** equilibrium.

Solution to 3: Let $\lambda = \frac{M}{L}$ be the linear mass density. When a length x has fallen, the mass resting on the scale is $m_{\text{rest}} = \lambda x$. The weight of this portion is $W = \lambda xg$. The remaining chain above is in free fall. The velocity of the chain just before hitting the scale is found from $v^2 = u^2 + 2gx = 2gx$. In a time dt , a small mass $dm = \lambda v dt$ hits the scale and comes to a complete stop. The force required to stop this falling mass is $F_{\text{impact}} = \frac{dp}{dt} = \frac{v dm}{dt} = v(\lambda v) = \lambda v^2$. Substitute $v^2 = 2gx$: $F_{\text{impact}} = \lambda(2gx) = 2\lambda xg$. The total reading on the scale is the sum of the static weight and the impact force: $F_{\text{total}} = W + F_{\text{impact}} = \lambda xg + 2\lambda xg = 3\lambda xg = 3\left(\frac{M}{L}\right)xg$. Fascinatingly, the scale reads exactly 3 times the weight of the chain that has accumulated on it!

Chapter 6

Gravitation

Newton's Law of Universal Gravitation was the first great synthesis in physics, uniting the motion of apples falling on Earth with the orbits of planets in the heavens.

6.1 Newton's Law of Gravitation

Every particle in the universe attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Key Formula: Law of Gravitation

$$\mathbf{F}_{12} = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}}$$

where $G \approx 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ is the universal gravitational constant, and $\hat{\mathbf{r}}$ is the unit vector pointing from mass 1 to mass 2. The negative sign indicates it is an attractive force.

6.2 Gravitational Field and Potential

Instead of thinking about forces acting instantaneously over vast distances, physicists use the concept of a **field**. A mass creates a gravitational field $\mathbf{g}$ in the space around it, and a second mass placed in that field experiences a force $\mathbf{F} = m\mathbf{g}$.

Definition: Gravitational Field

The gravitational field $\mathbf{g}$ at a point is the gravitational force experienced by a test mass m_0 divided by m_0 :

$$\mathbf{g} = \frac{\mathbf{F}}{m_0} = -\frac{GM}{r^2} \hat{\mathbf{r}}$$

Because gravity is a conservative force, we can define a **gravitational potential energy** U for a system of two masses. Choosing $U = 0$ at $r = \infty$:

$$U(r) = -\frac{Gm_1 m_2}{r}$$

The **gravitational potential** V is the potential energy per unit mass ($V = \frac{U}{m}$):

$$V(r) = -\frac{GM}{r}$$

The field is the negative gradient of the potential: $\mathbf{g} = -\nabla V$.

6.3 The Shell Theorem

Newton delayed publishing his Principia for years because he needed to prove that a spherical planet acts gravitationally as if all its mass were concentrated at its center. This is known as the Shell Theorem.

Theorem 1: A uniform spherical shell of matter exerts no net gravitational force on a particle located **inside** it.

Theorem 2: A uniform spherical shell of matter attracts a particle **outside** it as if all the shell's mass were concentrated at its center.

Example: Derivation of the Shell Theorem (Outside)

Consider a thin uniform spherical shell of mass M and radius R . A point mass m is located at a distance $r > R$ from the center.

We slice the shell into thin rings defined by an angle θ from the axis connecting the center of the shell to m . The radius of the ring is $R \sin \theta$, and its width is $R d\theta$.

The area of this ring is $dA = (2\pi R \sin \theta)(R d\theta)$. Since the shell is uniform, the mass of the ring is $dM = M \frac{dA}{4\pi R^2} = \frac{M}{2} \sin \theta d\theta$.

Every point on this ring is at the same distance s from the mass m . By the Law of Cosines:

$$s^2 = R^2 + r^2 - 2Rr \cos \theta$$

Differentiating implicitly with respect to θ (holding R and r constant):

$$2s ds = 2Rr \sin \theta d\theta \Rightarrow \sin \theta d\theta = \frac{s ds}{Rr}$$

Substituting this into dM :

$$dM = \frac{M}{2} \frac{s ds}{Rr}$$

The gravitational force dF exerted by the ring on m points entirely along the symmetry axis (transverse components cancel). The axial component is $\frac{r - R \cos \theta}{s}$. Using the Law of Cosines again, this factor is $\frac{s^2 + r^2 - R^2}{2rs}$.

The total force is the integral of $\frac{Gm dM}{s^2}$ multiplied by this directional factor:

$$\begin{aligned} F &= \int_{r-R}^{r+R} \frac{GmM}{2Rr} \left(\frac{s^2 + r^2 - R^2}{2rs^2} \right) ds \\ F &= \frac{GmM}{4Rr^2} \int_{r-R}^{r+R} \left(1 + \frac{r^2 - R^2}{s^2} \right) ds \\ F &= \frac{GmM}{4Rr^2} \left[s - \frac{r^2 - R^2}{s} \right]_{r-R}^{r+R} = \frac{GmM}{r^2} \end{aligned}$$

The shell acts exactly as a point mass M at its center!

6.4 Gravity Inside and Outside Planets

Using the Shell Theorem, we can determine the gravitational field g for a solid, uniform planet of radius R and mass M .

- **Outside ($r \geq R$):** The entire planet acts as a point mass. $g = \frac{GM}{r^2}$.
- **Inside ($r < R$):** Only the mass **inside** the radius r contributes to the gravity. The outer shells exert zero net force.

$$M_{\text{enclosed}} = M \frac{r^3}{R^3}$$

$$g_{\text{inside}} = \frac{GM_{\text{enclosed}}}{r^2} = \frac{GM}{R^3} r$$

Gravity inside a uniform planet increases **linearly** from the center to the surface!

6.5 Escape Velocity and Orbits

Definition: Escape Velocity

The minimum speed required for an object at the surface of a planet to escape its gravitational pull completely (reaching $r = \infty$ with $v = 0$).

Using conservation of energy:

$$\frac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{R} = 0 \Rightarrow v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

Geostationary Satellites: A satellite that appears stationary from Earth must orbit over the equator, rotating in the same direction as Earth, with an orbital period of exactly 24 hours. Equating centripetal force to gravitational force:

$$\frac{mv^2}{r} = \frac{GMm}{r^2} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

Since $v = \frac{2\pi r}{T}$, we find the orbital radius:

$$r^3 = \frac{GMT^2}{4\pi^2}$$

6.6 Kepler's Laws

Johannes Kepler deduced three empirical laws of planetary motion, which Newton later proved theoretically.

1. **Law of Orbits:** All planets move in elliptical orbits, with the Sun at one focus.
2. **Law of Areas:** A line that connects a planet to the Sun sweeps out equal areas in equal times. (This is a direct consequence of the conservation of angular momentum).
3. **Law of Periods:** The square of the period of any planet is proportional to the cube of the semimajor axis of its orbit.

$$T^2 = \frac{4\pi^2}{GM}a^3$$

Chapter Summary

- **Newton's Law of Gravitation** governs the attractive force between masses.
- **Potential Energy** $U = -Gm_1\frac{m_2}{r}$ is always negative, maximizing at zero at infinity.
- The **Shell Theorem** simplifies the analysis of spherical bodies. Inside a uniform sphere, g is proportional to r . Outside, it falls off as $\frac{1}{r^2}$.
- **Kepler's Second Law** derives from angular momentum conservation. The Third Law connects period and orbital radius.

6.6.1 Challenge Problems

1. **Beginner:** Calculate the escape velocity from the surface of the Moon. ($M_{\text{moon}} = 7.35 \times 10^{22}$ kg, $R_{\text{moon}} = 1737$ km).
2. **Intermediate:** Imagine a straight tunnel is drilled completely through the Earth, passing through its center. If a ball is dropped into the tunnel, show that it undergoes Simple Harmonic Motion (SHM) and find its period. (Assume uniform Earth density).
3. **Advanced (Olympiad):** A satellite is in a circular orbit of radius r . It is momentarily slowed down by a very short burst from its retrorockets, reducing its velocity by a small fraction Δv . Find the new perigee and apogee distances of its resulting elliptical orbit.

Chapter 7

Oscillation

Oscillatory motion is everywhere in nature: a swinging pendulum, a vibrating guitar string, atoms in a crystal lattice, and even alternating electric currents. When a system is displaced from a stable equilibrium, a restoring force attempts to pull it back, resulting in back-and-forth motion.

7.1 Simple Harmonic Motion (SHM)

Simple Harmonic Motion (SHM) is the most fundamental type of oscillation. It occurs whenever the restoring force is directly proportional to the displacement from equilibrium (Hooke's Law).

Definition: Conditions for SHM

A particle undergoes SHM if its acceleration is proportional to its displacement x and directed towards the equilibrium position ($x = 0$):

$$a = -\omega^2 x$$

where ω is a positive constant called the **angular frequency**.

Since $a = \frac{d^2x}{dt^2}$, this is a second-order linear differential equation:

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

The general solution is a sinusoidal function:

$$x(t) = A \cos(\omega t + \varphi)$$

- A : **Amplitude** (maximum displacement).
- $\omega = 2\pi f = \frac{2\pi}{T}$: **Angular frequency** (rad/s).
- φ : **Phase constant** (depends on initial conditions).
- $T = \frac{2\pi}{\omega}$: **Period** (time for one complete cycle).

Velocity and Acceleration in SHM: Differentiating $x(t)$ with respect to time:

$$v(t) = \frac{dx}{dt} = -A\omega \sin(\omega t + \varphi)$$

$$a(t) = \frac{dv}{dt} = -A\omega^2 \cos(\omega t + \varphi) = -\omega^2 x(t)$$

Maximum speed is $v_{\max} = A\omega$, occurring at the equilibrium position ($x = 0$). Maximum acceleration is $a_{\max} = A\omega^2$, occurring at the turning points ($x = \pm A$).

7.2 Energy in SHM

If a mass-spring system (where $F = -kx$) undergoes SHM, its angular frequency is $\omega = \sqrt{\frac{k}{m}}$.

Potential Energy: $U(t) = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2(\omega t + \varphi)$ **Kinetic Energy:** $K(t) = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2 \sin^2(\omega t + \varphi) = \frac{1}{2}kA^2 \sin^2(\omega t + \varphi)$

The total mechanical energy $E = K + U$ is constant:

$$E = \frac{1}{2}kA^2(\sin^2(\dots) + \cos^2(\dots)) = \frac{1}{2}kA^2$$

Energy oscillates continuously between pure kinetic at $x = 0$ and pure potential at $x = \pm A$.

7.3 Common SHM Systems**1. Mass on a Spring:**

$$T = 2\pi\sqrt{\frac{m}{k}}$$

2. Simple Pendulum: A mass m hanging from a massless string of length L . For small angles ($\sin \theta \approx \theta$):

$$\tau = -mgL \sin \theta \approx -mgL\theta = I\alpha = (mL^2)\alpha$$

$$\alpha = -\frac{g}{L}\theta \Rightarrow \omega = \sqrt{\frac{g}{L}} \Rightarrow T = 2\pi\sqrt{\frac{L}{g}}$$

3. Physical Pendulum: A rigid body of mass m and moment of inertia I pivoted at a distance d from its center of mass.

$$\tau = -mgd \sin \theta \approx -mgd\theta = I\alpha$$

$$\omega = \sqrt{\frac{mgd}{I}} \Rightarrow T = 2\pi\sqrt{\frac{I}{mgd}}$$

7.4 Damped Oscillation

In reality, friction and air resistance eventually bring oscillators to rest. A common model for damping is a force proportional to velocity: $F_d = -bv$, where b is the damping constant.

The equation of motion becomes:

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

For **underdamped** systems (small b), the solution is an oscillation with an exponentially decaying amplitude:

$$x(t) = Ae^{-\frac{bt}{2m}} \cos(\omega' t + \varphi)$$

where the damped angular frequency is $\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$.

Olympiad Tip: Types of Damping

- **Underdamped:** System oscillates with decreasing amplitude.
- **Critically Damped:** System returns to equilibrium as fast as possible without oscillating ($b^2 = 4mk$).
- **Overdamped:** System slowly creeps back to equilibrium without oscillating ($b^2 > 4mk$).

7.5 Forced Oscillation and Resonance

If an external driving force $F_{\text{ext}} = F_0 \cos(\omega_d t)$ is applied to a damped oscillator, it is forced to oscillate at the driving frequency ω_d .

After the initial transient behavior dies out, the steady-state amplitude is:

$$A = \frac{F_0}{\sqrt{m^2(\omega_d^2 - \omega^2)^2 + b^2\omega_d^2}}$$

where $\omega = \sqrt{\frac{k}{m}}$ is the natural frequency of the undamped system.

Key Formula: Resonance

When the driving frequency ω_d matches the natural frequency ω , the amplitude A reaches a dramatic maximum. This phenomenon is called **resonance**. At resonance ($\omega_d = \omega$), $A = \frac{F_0}{b\omega}$.

Chapter Summary

- **SHM** occurs when a restoring force is proportional to displacement ($a = -\omega^2 x$).
- The solution is sinusoidal: $x(t) = A \cos(\omega t + \varphi)$.
- **Energy** in SHM is proportional to the square of the amplitude ($E = \frac{1}{2}kA^2$).
- **Damping** causes the amplitude to decay exponentially over time.
- **Resonance** occurs when a driving force matches the system's natural frequency, resulting in large-amplitude oscillations.

7.5.1 Challenge Problems

1. **Beginner:** A 2 kg mass is attached to a spring with $k = 50$ N/m. It is displaced 10 cm and released. Find its maximum speed.
2. **Intermediate:** A uniform solid cylinder of mass M and radius R is attached to a horizontal spring of constant k . It rolls without slipping. Show that it undergoes SHM and find its period. (Hint: Include rotational kinetic energy).

3. **Advanced (Olympiad):** A small block rests on a horizontal board. The board is driven in horizontal SHM with amplitude A and frequency f . If the coefficient of static friction is μ_s , what is the maximum amplitude A before the block begins to slip?

Chapter 8

Fluids

A fluid is any substance that can flow and alter its shape to the container it is in. Both liquids and gases are fluids. While liquids are largely incompressible, gases are highly compressible.

8.1 Density and Pressure

Density (ρ): The mass per unit volume of a substance.

$$\rho = \frac{m}{V}$$

The SI unit is kg/m^3 . For water, $\rho \approx 1000 \text{ kg/m}^3$.

Pressure (P): The normal force exerted by a fluid per unit area.

$$P = \frac{F_{\perp}}{A}$$

The SI unit is the Pascal ($\text{Pa} = \text{N/m}^2$). Standard atmospheric pressure is $1 \text{ atm} \approx 1.013 \times 10^5 \text{ Pa}$.

8.2 Fluid Statics

Key Formula: Hydrostatic Pressure

The pressure at a depth h below the surface of a static, incompressible fluid is:

$$P = P_0 + \rho gh$$

where P_0 is the pressure at the surface (often atmospheric pressure).

This equation shows that pressure in a static fluid depends **only** on depth, not on the shape of the container.

Pascal's Principle: A change in pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and to the walls of its container. This is the underlying principle behind hydraulic lifts.

Archimedes' Principle: Any object, wholly or partially immersed in a fluid, is buoyed up by a force equal to the weight of the fluid displaced by the object.

$$F_b = W_{\text{displaced fluid}} = \rho_{\text{fluid}} V_{\text{submerged}} g$$

If $\rho_{\text{object}} < \rho_{\text{fluid}}$, the object floats. If $\rho_{\text{object}} > \rho_{\text{fluid}}$, the object sinks.

8.3 Fluid Dynamics

For ideal fluids (incompressible, non-viscous, steady flow, and irrotational), we can analyze their motion using two powerful equations.

Key Formula: Equation of Continuity

For an incompressible fluid flowing through a pipe of varying cross-section, the volume flow rate ($Q = Av$) must be constant:

$$A_1 v_1 = A_2 v_2$$

This implies that a fluid must flow faster through a narrow section of a pipe than through a wider section.

Key Formula: Bernoulli's Equation

Bernoulli's equation is a statement of the conservation of energy per unit volume for a flowing ideal fluid:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 = \text{constant}$$

Consequences of Bernoulli's Equation:

- **Torricelli's Law:** The speed of efflux of a fluid from a small hole at depth h in an open tank is $v = \sqrt{2gh}$.
- **Venturi Effect:** Where velocity is high, pressure is low. This explains how airplane wings generate lift and how carburetors work.

8.4 Viscosity and Poiseuille's Law

Real fluids have internal friction called **viscosity** (η). Viscosity resists the relative motion of adjacent layers of fluid.

For laminar (smooth) flow of a viscous fluid through a cylindrical pipe of radius R and length L , the volume flow rate Q is given by **Poiseuille's Law**:

$$Q = \frac{\pi R^4 (P_1 - P_2)}{8\eta L}$$

Notice the R^4 dependence! A small decrease in the radius of a pipe (or a blood vessel) requires a massive increase in pressure difference to maintain the same flow rate.

8.5 Surface Tension

Molecules at the surface of a liquid experience a net inward cohesive force, creating a surface that behaves like a stretched elastic membrane. **Surface tension** (γ or T) is the force per unit length acting along the surface.

$$\gamma = \frac{F}{L}$$

Surface tension is responsible for the spherical shape of drops and the capillary action that draws water up narrow tubes.

Key Formula: Capillary Rise

The height h that a liquid rises in a capillary tube of radius r is:

$$h = \frac{2\gamma \cos \theta}{\rho g r}$$

where θ is the contact angle between the liquid and the tube wall.

Chapter Summary

- **Hydrostatic pressure** increases linearly with depth: $P = P_0 + \rho g h$.
- **Archimedes' principle** dictates that buoyant force equals the weight of displaced fluid.
- The **Continuity equation** ($A_1 v_1 = A_2 v_2$) conserves mass.
- **Bernoulli's equation** relates pressure, speed, and height, reflecting energy conservation.
- **Viscosity** dictates flow rate in narrow tubes via Poiseuille's Law ($Q \propto R^4$).

8.5.1 Challenge Problems

1. **Beginner:** A block of wood ($\rho = 600 \text{ kg/m}^3$) floats in water ($\rho = 1000 \text{ kg/m}^3$). What fraction of its volume is submerged?
2. **Intermediate:** Water flows continuously from a faucet of radius R_0 . As the water falls, the stream narrows. Assuming the water falls from rest, find the radius R of the stream at a distance h below the faucet.

3. **Advanced (Olympiad):** A U-tube of uniform cross-section contains two liquids of densities ρ_1 and ρ_2 ($\rho_1 > \rho_2$) that do not mix. If the system is slightly displaced from equilibrium and released, it oscillates. Derive the period of oscillation.

Chapter 9

Waves & Acoustics

A wave is a disturbance that propagates through space and time, transferring energy and momentum without the bulk transport of matter. Acoustics is the branch of physics that studies sound, which is a mechanical wave.

9.1 Types of Waves

Waves are classified by how the medium oscillates relative to the direction of propagation.

1. **Transverse Waves:** The particles of the medium oscillate perpendicular to the direction of wave travel. Examples: Waves on a string, electromagnetic waves (light).
2. **Longitudinal Waves:** The particles of the medium oscillate parallel to the direction of wave travel, creating alternating regions of compression and rarefaction. Example: Sound waves in a gas or liquid.

9.2 The Wave Equation and Velocity

A traveling harmonic wave moving in the positive x -direction is described by the wavefunction:

$$y(x, t) = A \cos(kx - \omega t + \varphi)$$

- A : Amplitude.
- $k = \frac{2\pi}{\lambda}$: Wavenumber, where λ is the wavelength.
- $\omega = \frac{2\pi}{T} = 2\pi f$: Angular frequency.
- φ : Phase constant.

The phase velocity v of the wave is the speed at which a crest travels:

$$v = f\lambda = \frac{\omega}{k}$$

Wave Velocity in Different Media:

- Transverse wave on a stretched string: $v = \sqrt{\frac{T}{\mu}}$, where T is tension and μ is linear mass density.
- Longitudinal sound wave in a fluid: $v = \sqrt{\frac{B}{\rho}}$, where B is the bulk modulus.
- Sound in an ideal gas: $v = \sqrt{\frac{\gamma RT}{M}}$, where γ is the adiabatic index, R is the gas constant, T is temperature, and M is molar mass.

9.3 Superposition and Interference

The **Principle of Superposition** states that when two or more waves overlap, the net displacement is the algebraic sum of their individual displacements.

When two harmonic waves of the same frequency and amplitude travel in the same direction but with a phase difference $\Delta\varphi$, their superposition gives:

$$y_1 + y_2 = A \cos(kx - \omega t) + A \cos(kx - \omega t + \Delta\varphi)$$

$$y_{\text{total}} = \left(2A \cos\left(\frac{\Delta\varphi}{2}\right)\right) \cos\left(kx - \omega t + \frac{\Delta\varphi}{2}\right)$$

- **Constructive Interference:** Occurs when $\Delta\varphi = 2\pi m$ (or path difference $\Delta x = m\lambda$). The resultant amplitude is $2A$.
- **Destructive Interference:** Occurs when $\Delta\varphi = 2\pi\left(m + \frac{1}{2}\right)$ (or $\Delta x = \left(m + \frac{1}{2}\right)\lambda$). The resultant amplitude is 0.

9.4 Stationary (Standing) Waves

When two identical waves travel in opposite directions, their superposition creates a standing wave.

$$y(x, t) = A \sin(kx - \omega t) + A \sin(kx + \omega t) = (2A \sin kx) \cos \omega t$$

The term $(2A \sin kx)$ is the spatial amplitude.

- **Nodes:** Points of zero amplitude ($kx = m\pi \Rightarrow x = m\frac{\lambda}{2}$).
- **Antinodes:** Points of maximum amplitude ($x = (m + \frac{1}{2})\frac{\lambda}{2}$).

For a string of length L fixed at both ends, the boundary conditions require nodes at $x = 0$ and $x = L$. This quantizes the allowed wavelengths:

$$L = n\frac{\lambda}{2} \Rightarrow \lambda_n = \frac{2L}{n}$$

The corresponding resonant frequencies are $f_n = n\frac{v}{2L}$ for $n = 1, 2, 3, \dots$

9.5 Beats

When two waves of slightly different frequencies (f_1 and f_2) interfere, the resulting amplitude slowly modulates over time. The “loud-soft” pattern is called beats.

Key Formula: Beat Frequency

The frequency of the amplitude envelope (the number of beats per second) is:

$$f_{\text{beat}} = |f_1 - f_2|$$

9.6 The Doppler Effect

The Doppler effect is the perceived change in frequency of a wave when the source and observer are in relative motion.

Detailed Derivation for Sound: Let v be the speed of sound, v_s the speed of the source, and v_o the speed of the observer. Both speeds are measured relative to the medium (air).

1. **Moving Source:** If the source moves toward the observer, the waves are compressed. In one period $T = \frac{1}{f_s}$, the wave travels distance vT , and the source travels v_sT . The effective wavelength is:

$$\lambda' = vT - v_sT = \frac{v - v_s}{f_s}$$

2. **Moving Observer:** If the observer moves toward the source, they encounter wave crests faster. Their relative velocity to the wave is $v + v_o$.
3. **Combined Effect:** The perceived frequency f_o is the relative velocity divided by the effective wavelength:

$$f_o = \frac{v + v_o}{\lambda'} = f_s \frac{v + v_o}{v - v_s}$$

Key Formula: Doppler Effect General Formula

$$f_o = f_s \frac{v \pm v_o}{v \mp v_s}$$

- Use upper signs (numerator +, denominator −) for **approaching**.
- Use lower signs (numerator −, denominator +) for **receding**.

9.7 Diffraction

Diffraction is the bending and spreading of a wave as it passes through an aperture or around an obstacle. It becomes significant when the size of the aperture is comparable to the wavelength λ . For a single slit of width a , the first minimum in the diffraction pattern occurs at an angle θ where $a \sin \theta = \lambda$.

Chapter Summary

- **Wave velocity** depends on the medium’s elasticity and inertia.
- **Interference** depends on the phase difference between overlapping waves.
- **Standing waves** form from oppositely traveling identical waves, trapping energy between nodes.

- **Beats** arise from the interference of slightly different frequencies.
- The **Doppler Effect** changes the perceived frequency due to relative motion between source and observer.

9.7.1 Challenge Problems

1. **Beginner:** A siren emits a 1000 Hz sound. It moves towards a stationary observer at 34 m/s. If the speed of sound is 340 m/s, what frequency does the observer hear?
2. **Intermediate:** An organ pipe open at both ends has a fundamental frequency of 300 Hz. If one end is suddenly closed, what is the new fundamental frequency?
3. **Advanced (Olympiad):** Two speakers separated by distance d emit coherent sound waves of wavelength λ in phase. An observer walks along a line perpendicular to the axis connecting the speakers, starting from the midpoint. Derive an expression for the positions of the interference minima.

Chapter 10

Heat & Thermodynamics

Thermodynamics is the study of heat, work, temperature, and the statistical behavior of multi-particle systems. It bridges the microscopic world of bouncing atoms (Kinetic Theory) with the macroscopic world of engines and refrigerators.

10.1 Temperature and Thermal Expansion

Temperature is a measure of the average random translational kinetic energy of molecules. When the temperature of a substance increases, the increased kinetic energy causes molecules to push further apart, resulting in **thermal expansion**.

Key Formula: Linear and Volume Expansion

For solids, the change in length is proportional to the original length and the temperature change:

$$\Delta L = \alpha L_0 \Delta T$$

where α is the coefficient of linear expansion.

For volumes (solids or liquids):

$$\Delta V = \beta V_0 \Delta T$$

where $\beta \approx 3\alpha$ is the coefficient of volume expansion.

10.2 Heat and Calorimetry

Heat (Q) is the transfer of energy due to a temperature difference.

- **Specific Heat Capacity (c):** The heat required to raise the temperature of 1 kg of a substance by 1 K.

$$Q = mc\Delta T$$

- **Latent Heat (L):** The heat required to change the phase of 1 kg of a substance **without** changing its temperature.

$$Q = mL$$

(Use L_f for fusion/melting and L_v for vaporization/boiling).

In an isolated system, the principle of calorimetry dictates that heat lost by hot objects equals heat gained by cold objects: $\sum Q = 0$.

10.3 The Ideal Gas Law and Kinetic Theory

An ideal gas is a theoretical gas composed of many randomly moving point particles that do not interact except in perfectly elastic collisions.

Key Formula: Ideal Gas Law

$$PV = nRT = Nk_B T$$

- P : Pressure, V : Volume, T : Absolute temperature (Kelvin)
- n : Number of moles, $R = 8.314 \text{ J/(mol}\cdot\text{K)}$: Universal gas constant
- N : Number of molecules, $k_B = 1.38 \times 10^{-23} \text{ J/K}$: Boltzmann constant

Kinetic Theory of Gases: The macroscopic pressure of a gas is the result of microscopic molecular collisions with the container walls.

$$P = \frac{1}{3} \frac{Nm}{V} v_{\text{rms}}^2$$

The root-mean-square speed v_{rms} is directly related to temperature:

$$\frac{1}{2} m v_{\text{rms}}^2 = \frac{3}{2} k_B T$$

This shows that temperature is exactly proportional to the average kinetic energy of the molecules.

10.4 The First Law of Thermodynamics

The First Law is simply the conservation of energy applied to a thermodynamic system.

Key Formula: First Law of Thermodynamics

$$\Delta E_{\text{int}} = Q - W$$

- ΔE_{int} : Change in internal energy (depends **only** on temperature for an ideal gas).
- Q : Heat added **to** the system.
- W : Work done **by** the system ($W = \int P dV$).

Thermodynamic Processes:

1. **Isothermal:** Constant temperature ($\Delta T = 0 \Rightarrow \Delta E_{\text{int}} = 0$). Work done is $W = nRT \ln\left(\frac{V_f}{V_i}\right)$.
2. **Isobaric:** Constant pressure ($\Delta P = 0$). Work done is $W = P\Delta V$.
3. **Isochoric (Isovolumetric):** Constant volume ($\Delta V = 0 \Rightarrow W = 0$). All heat goes into internal energy.
4. **Adiabatic:** No heat exchange ($Q = 0$). Process follows $PV^\gamma = \text{constant}$, where $\gamma = \frac{C_p}{C_v}$ is the ratio of specific heats.

10.5 Heat Engines and the Second Law

A heat engine takes heat Q_H from a hot reservoir at T_H , converts a portion into useful work W , and exhausts the remaining heat Q_C into a cold reservoir at T_C . The efficiency is $e = \frac{W}{Q_H} = \frac{Q_H - |Q_C|}{Q_H} = 1 - \frac{|Q_C|}{Q_H}$.

The Second Law of Thermodynamics states that heat flows spontaneously from hot to cold, and no heat engine can be 100% efficient.

Carnot Engine: The theoretical maximum efficiency for any engine operating between T_H and T_C is achieved by a reversible Carnot cycle:

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_H}$$

(Temperatures **must** be in Kelvin).

10.6 Entropy

Entropy (S) is a measure of the disorder, or multiplicity of microstates, in a system. Macroscopically, the change in entropy for a reversible process is:

$$\Delta S = \int \frac{dQ_{\text{rev}}}{T}$$

The Second Law can be restated: The total entropy of an isolated system (or the Universe) can never decrease. $\Delta S_{\text{univ}} \geq 0$.

Chapter Summary

- **Thermal expansion** occurs because atoms vibrate with greater amplitude at higher temperatures.
- **Phase changes** require latent heat and occur at constant temperature.
- The **First Law** ($\Delta E_{\text{int}} = Q - W$) is energy conservation.
- **Adiabatic processes** have no heat exchange ($Q = 0$); **Isothermal** processes have no internal energy change ($\Delta E_{\text{int}} = 0$).
- The **Second Law** forbids 100% efficient engines and dictates that entropy universally increases.

10.6.1 Challenge Problems

1. **Beginner:** A gas expands from 1.0 m^3 to 3.0 m^3 at a constant pressure of 200 kPa. How much work is done by the gas?
2. **Intermediate:** An ideal monatomic gas undergoes an adiabatic expansion to twice its original volume. If its initial temperature was 300 K, what is its final temperature? (For a monatomic gas, $\gamma = \frac{5}{3}$).
3. **Advanced (Olympiad):** A Carnot engine operates between a hot reservoir at T_H and a cold reservoir at T_C . The engine drives a Carnot refrigerator operating between the same two reservoirs. If the engine extracts heat Q_1 from the hot reservoir, what is the net heat extracted from the hot reservoir by the combined system?

Chapter 11

Optics

Optics is the study of light and its interactions with matter. It is broadly divided into Geometric (Ray) Optics, where light travels in straight lines, and Physical (Wave) Optics, where interference and diffraction effects become prominent.

11.1 Geometric Optics: Reflection and Refraction

Light travels through a vacuum at speed $c \approx 3.0 \times 10^8 \text{ m/s}$. In a material medium, it slows down. The **index of refraction** (n) is defined as:

$$n = \frac{c}{v}$$

Since $v \leq c$, the index of refraction is always $n \geq 1$. (For air, $n \approx 1$).

Law of Reflection: When a light ray strikes a smooth surface, the angle of incidence equals the angle of reflection:

$$\theta_i = \theta_r$$

(All angles are measured relative to the **normal**, the line perpendicular to the surface).

Snell's Law of Refraction: When light crosses a boundary between two media, its speed changes, causing it to bend.

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

If light enters a **denser** medium ($n_2 > n_1$), it bends **towards** the normal. If light enters a **less dense** medium ($n_2 < n_1$), it bends **away from** the normal.

Key Formula: Total Internal Reflection

When light attempts to move from a denser to a less dense medium ($n_1 > n_2$), there is a critical angle θ_c above which no light is refracted; it is entirely reflected back.

$$\sin \theta_c = \frac{n_2}{n_1}$$

11.2 Lenses and Mirrors

The formation of images by curved mirrors and thin lenses is governed by standard equations using a sign convention (Cartesian or Real-is-Positive).

The Mirror / Thin Lens Equation:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

- u : Object distance (usually positive).
- v : Image distance (positive for real images, negative for virtual images).
- f : Focal length (positive for converging lenses/mirrors, negative for diverging).

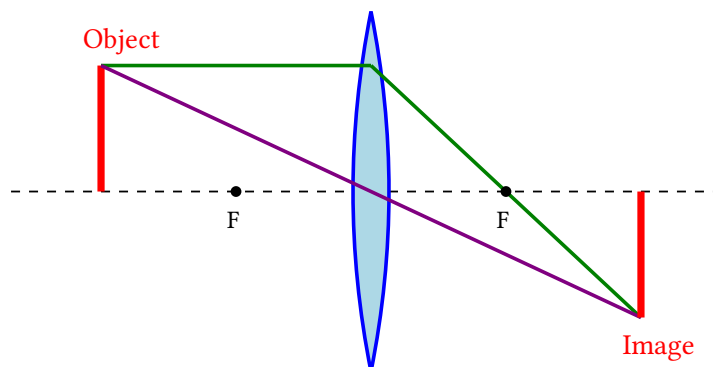
Magnification:

$$m = -\frac{v}{u} = \frac{h_i}{h_o}$$

If m is negative, the image is inverted. If $|m| > 1$, the image is magnified.

Lens Maker's Equation: For a thin lens in air with index n and radii of curvature R_1 and R_2 :

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$



11.3 Prisms and Dispersion

Because the index of refraction n depends slightly on wavelength, different colors of light bend by different amounts. This is called **dispersion**. Blue light (shorter wavelength) bends **more** than red light.

When a monochromatic ray passes through a prism of refracting angle A , it undergoes a total deviation δ .

$$\delta = \theta_i + \theta_e - A$$

Minimum deviation (δ_m) occurs when the ray passes symmetrically through the prism ($\theta_i = \theta_e$).

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

11.4 Physical Optics: Interference

Geometric optics assumes light travels in perfect straight lines. Physical optics accounts for the wave nature of light.

Young's Double-Slit Experiment: Coherent light of wavelength λ illuminates two narrow slits separated by a distance d . The waves interfere on a screen at distance L . The path difference between the two waves is $\Delta x = d \sin \theta$.

- **Constructive interference (bright fringes):** $d \sin \theta = m\lambda$
- **Destructive interference (dark fringes):** $d \sin \theta = (m + \frac{1}{2})\lambda$

For small angles ($\sin \theta \approx \frac{y}{L}$), the position of the m -th bright fringe on the screen is $y_m = \frac{m\lambda L}{d}$.

11.5 Physical Optics: Diffraction

Single-Slit Diffraction: When light passes through a single narrow slit of width a , it spreads out. The path difference between a ray from the top of the slit and a ray from the middle is $(\frac{a}{2}) \sin \theta$. When this path difference is $\frac{\lambda}{2}$, the rays destructively interfere.

- **Dark fringes (minima):** $a \sin \theta = m\lambda$ (for $m = \pm 1, \pm 2, \dots$)

Note that $m = 0$ is the central **maximum**, which is twice as wide as the other maxima.

Chapter Summary

- **Snell's Law** ($n_1 \sin \theta_1 = n_2 \sin \theta_2$) governs refraction.
- **Total Internal Reflection** occurs when $\theta_i > \theta_c$ going from high n to low n .
- The **Thin Lens/Mirror Equation** ($\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$) locates images.
- **Young's Double-Slit** produces evenly spaced bright and dark interference fringes.
- **Single-Slit Diffraction** creates a wide central maximum flanked by narrower, dimmer maxima.

11.5.1 Challenge Problems

1. **Beginner:** An object is placed 15 cm in front of a convex lens of focal length 10 cm. Find the image position and magnification. Is it real or virtual?
2. **Intermediate:** A light ray enters a glass fiber ($n = 1.50$) from air. What is the maximum angle of incidence on the flat end of the fiber such that the ray undergoes total internal reflection on the cylindrical walls?
3. **Advanced (Olympiad):** In a Young's double slit setup, a thin glass plate of thickness t and index n is placed in front of one of the slits. Derive the expression for the lateral shift of the central maximum on the screen.

11.5.2 Fully Worked Solutions

Solution to 1: Given $u = 15$ cm (object distance is positive) and $f = +10$ cm (convex lens). Using the thin lens equation: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{10} = \frac{1}{v} + \frac{1}{15} \cdot \frac{1}{v} = \frac{1}{10} - \frac{1}{15} = \frac{3}{30} - \frac{2}{30} = \frac{1}{30}$. Thus, $v = +30$ cm. Since v is positive, the image is **real** and forms on the opposite side of the lens. Magnification $m = -\frac{v}{u} = -\frac{30}{15} = -2$. The image is real, inverted, and twice as large.

Solution to 2: Let the ray enter the flat end at angle θ_i and refract at angle θ_r . By Snell's law at the flat end: $\sin \theta_i = n \sin \theta_r$ (since $n_{\text{air}} = 1$). The ray hits the cylindrical wall at an angle of incidence θ_w relative to the normal of the wall. From geometry, $\theta_w = 90^\circ - \theta_r$. For total internal reflection at the wall, we need $\theta_w \geq \theta_c$, where $\sin \theta_c = \frac{1}{n}$. So, $\sin(90^\circ - \theta_r) \geq \frac{1}{n} \Rightarrow \cos \theta_r \geq \frac{1}{n}$. We know $\cos \theta_r = \sqrt{1 - \sin^2 \theta_r} = \sqrt{1 - \frac{\sin^2 \theta_i}{n^2}}$. Thus, $\sqrt{1 - \frac{\sin^2 \theta_i}{n^2}} \geq \frac{1}{n}$. Squaring both sides: $1 - \frac{\sin^2 \theta_i}{n^2} \geq \frac{1}{n^2} \cdot n^2 - \sin^2 \theta_i \geq 1 \Rightarrow \sin^2 \theta_i \leq n^2 - 1$. For $n = 1.50$, $n^2 - 1 = 2.25 - 1 = 1.25$. Since $\sin^2 \theta_i$ cannot exceed 1, the condition $\sin^2 \theta_i \leq 1.25$ is satisfied for **any** possible angle of incidence θ_i (0 to 90°). Thus, the maximum angle is 90° . All entering rays will be internally reflected!

Solution to 3: Without the glass plate, the central maximum occurs at $\theta = 0$ where the path difference is zero. When a glass plate of thickness t and index n is placed in one path, the light travels a distance t through glass instead of vacuum. The optical path length increases by $\Delta x_{\text{optical}} = nt - 1t = (n - 1)t$. To restore a net path difference of zero at the screen, the other ray must travel an extra geometric distance $(n - 1)t$. From the standard double slit geometry, the geometric path difference for a point at position y on the screen is $d \sin \theta \approx d(\frac{y}{L})$. Equating the two: $d(\frac{y}{L}) = (n - 1)t$. The lateral shift of the central maximum is $y = (\frac{L}{d})(n - 1)t$. The entire interference pattern shifts upward (towards the slit with the glass plate) by this amount.

Chapter 12

Electricity

Electricity is the study of electric charges, their interactions, and their movement. It forms one half of electromagnetism, arguably the most successful theory in classical physics.

12.1 Electrostatics: Force and Field

Electric charge comes in two types: positive and negative. Like charges repel, and opposite charges attract. Charge is quantized (in units of $e = 1.6 \times 10^{-19}$ C) and conserved.

Key Formula: Coulomb's Law

The electrostatic force between two point charges q_1 and q_2 separated by distance r is:

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}$$

where $\epsilon_0 \approx 8.85 \times 10^{-12}$ C²/(N·m²) is the vacuum permittivity, and $\frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9$ N·m²/C².

Electric Field ($\mathbf{E}$): Instead of charges interacting instantaneously across space, a charge creates an electric field $\mathbf{E}$ everywhere, which then exerts a force on a second test charge q_0 .

$$\mathbf{E} = \frac{\mathbf{F}}{q_0}$$

For a point charge q : $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$.

12.2 Gauss's Law

Gauss's Law is a powerful alternative to Coulomb's law for calculating electric fields of highly symmetric charge distributions.

Key Formula: Gauss's Law

The electric flux Φ_E through any closed surface is proportional to the total enclosed charge:

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

Applications:

- **Uniform sphere** of charge Q and radius R : Outside, it acts like a point charge ($E \propto \frac{1}{r^2}$). Inside, $E \propto r$.
- **Infinite line** of charge with linear density λ : $E = \frac{\lambda}{2\pi\epsilon_0 r}$.
- **Infinite sheet** of charge with surface density σ : $E = \frac{\sigma}{2\epsilon_0}$ (constant!).

12.3 Electric Potential

Since the electrostatic force is conservative, we can define an electric potential energy U . **Electric Potential (V)** is the potential energy per unit charge: $V = \frac{U}{q_0}$.

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{s}$$

For a point charge: $V(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$.

The electric field points in the direction of the steepest decrease in potential:

$$\mathbf{E} = -\nabla V$$

Common Misconception: Field vs. Potential

A region can have $V = 0$ but $\mathbf{E} \neq 0$ (e.g., exactly midway between a $+q$ and $-q$ dipole). Conversely, a region can have $\mathbf{E} = 0$ but $V \neq 0$ (e.g., inside a charged conducting sphere, where V is constant but non-zero).

12.4 Capacitance and Dielectrics

A capacitor is a device that stores electrical energy in an electric field, typically consisting of two isolated conductors.

$$C = \frac{Q}{V}$$

where C is capacitance (measured in Farads, F), Q is the charge on one plate, and V is the voltage difference between plates.

For a parallel-plate capacitor of area A and separation d :

$$C = \frac{\epsilon_0 A}{d}$$

The energy stored in a capacitor is:

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV$$

Dielectrics: Inserting an insulating material (dielectric) with dielectric constant $\kappa > 1$ between the plates increases the capacitance to $C = \kappa C_0$. The dielectric molecules polarize, partially canceling the internal electric field.

12.5 Current, Resistance, and DC Circuits

When charges are in motion, we have an electric current I .

$$I = \frac{dq}{dt}$$

Ohm's Law: For many materials, the current density $\mathbf{J}$ is proportional to the electric field: $\mathbf{J} = \sigma \mathbf{E}$, where σ is conductivity. Macroscopically, this is:

$$V = IR$$

Resistance and Resistivity: The resistance R of a wire depends on its geometry and the material's resistivity $\rho = \frac{1}{\sigma}$.

$$R = \rho \frac{L}{A}$$

12.6 Kirchhoff's Circuit Laws

Complex circuits cannot always be reduced to simple series/parallel resistor combinations. They must be solved using Kirchhoff's laws, which are statements of fundamental conservation principles.

1. **Kirchhoff's Current Law (KCL):** The sum of currents entering a junction equals the sum of currents leaving. (Conservation of charge).

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

2. **Kirchhoff's Voltage Law (KVL):** The directed sum of potential differences around any closed loop in a circuit is zero. (Conservation of energy).

$$\sum \Delta V = 0$$

Combinations:

- **Resistors:** $R_{\text{eq}} = R_1 + R_2$ (Series), $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$ (Parallel).
- **Capacitors:** $\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2}$ (Series), $C_{\text{eq}} = C_1 + C_2$ (Parallel). Note that the rules are reversed!

Chapter Summary

- **Coulomb's Law** dictates forces between charges; **Gauss's Law** relates flux to enclosed charge.
- **Electric Potential** is the energy per unit charge; **Field** is its negative gradient.
- **Capacitors** store energy in electric fields; dielectrics increase capacitance.
- **Ohm's Law** relates voltage, current, and resistance.
- **Kirchhoff's Laws** are based on the conservation of charge (junction) and energy (loop).

12.6.1 Challenge Problems

1. **Beginner:** Two identical conducting spheres, one with charge $+4\mu C$ and the other with $-2\mu C$, are touched together and then separated. What is the final charge on each?
2. **Intermediate:** Find the equivalent resistance of an infinite ladder of resistors, where the series resistors are R_1 and the parallel rungs are R_2 .
3. **Advanced (Olympiad):** A solid insulating sphere of radius R has a non-uniform volume charge density $\rho(r) = \rho_0\left(\frac{r}{R}\right)$. Find the electric field $E(r)$ both inside ($r < R$) and outside ($r > R$) the sphere.

Chapter 13

Magnetism

Magnetism is intimately connected to electricity. While electric fields are created by stationary charges, magnetic fields are created by **moving** charges (currents).

13.1 The Magnetic Field and Lorentz Force

A magnetic field $\mathbf{B}$ (measured in Tesla, T) exerts a force on a moving charge. The force is always perpendicular to both the velocity of the charge and the magnetic field.

Key Formula: Lorentz Force Law

The magnetic force on a point charge q moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ is:

$$\mathbf{F}_B = q(\mathbf{v} \times \mathbf{B})$$

The total electromagnetic force is the Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Because $\mathbf{F}_B$ is always perpendicular to $\mathbf{v}$, the magnetic field **cannot do work** on a free charge. It can only change the direction of motion, not the speed. A charged particle moving perpendicular to a uniform magnetic field will undergo uniform circular motion with a radius (cyclotron radius) of $r = \frac{mv}{qB}$.

For a straight wire of length L carrying a steady current I in a uniform magnetic field:

$$\mathbf{F}_B = I(\mathbf{L} \times \mathbf{B})$$

13.2 Sources of Magnetic Field

How do currents create magnetic fields?

Biot-Savart Law: The fundamental law relating a differential current element $I d\mathbf{s}$ to the differential magnetic field $d\mathbf{B}$ it produces at a position vector $\mathbf{r}$ is:

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{d\mathbf{s} \times \hat{\mathbf{r}}}{r^2}$$

where $\mu_0 \approx 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ is the vacuum permeability.

Ampere's Law: For highly symmetric current distributions, Ampere's Law is a much simpler alternative to the Biot-Savart law.

$$\oint \mathbf{B} \cdot d\mathbf{s} = \mu_0 I_{\text{enclosed}}$$

Using Ampere's Law, we can easily find:

- **Infinite straight wire:** $B = \frac{\mu_0 I}{2\pi r}$
- **Inside a long solenoid:** $B = \mu_0 n I$ (where n is turns per unit length).

13.3 Electromagnetic Induction

In 1831, Michael Faraday discovered that a changing magnetic field induces an electric current in a nearby circuit.

Definition: Magnetic Flux

The magnetic flux Φ_B through an area A is the dot product of the magnetic field and the area vector:

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

Key Formula: Faraday's Law of Induction

The induced electromotive force ($\mathcal{E}$) in any closed loop is equal to the negative of the time rate of change of the magnetic flux enclosed by the loop.

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Lenz's Law (The Negative Sign): The induced current will flow in such a direction that its own magnetic field **opposes** the change in the original magnetic flux that produced it. (This is a consequence of energy conservation).

13.4 Inductance

A changing current in a coil creates a changing magnetic field, which in turn induces an EMF in the coil itself. This is called **self-inductance** (L).

$$\mathcal{E} = -L \frac{dI}{dt}$$

The unit of inductance is the Henry (H). An inductor resists **changes** in current. The energy stored in the magnetic field of an inductor is $U = \frac{1}{2} L I^2$.

Mutual Inductance (M): When a changing current in one coil induces an EMF in a nearby second coil:

$$\mathcal{E}_2 = -M \frac{dI_1}{dt}$$

13.5 Alternating Current and Transformers

If a coil is rotated at a constant angular velocity ω in a uniform magnetic field, the flux is $\Phi_B = BA \cos(\omega t)$. By Faraday's Law, the induced EMF is sinusoidal: $\mathcal{E} = \mathcal{E}_0 \sin(\omega t)$. This is the basis of an AC generator.

Transformers: A transformer consists of two coils (Primary and Secondary) wrapped around a common iron core. It uses mutual induction to step up or step down AC voltages.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

where N is the number of turns. Assuming 100% efficiency ($P_{\text{in}} = P_{\text{out}}$):

$$\frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Chapter Summary

- **Lorentz Force** ($\mathbf{F} = q\mathbf{v} \times \mathbf{B}$) causes moving charges to curve.
- **Ampere's Law** relates the line integral of $\mathbf{B}$ to the enclosed current.
- **Faraday's Law** states that a changing magnetic flux induces an EMF.
- **Lenz's Law** determines the direction of the induced current (it opposes the change).
- **Inductors** store energy in magnetic fields and oppose changes in current.
- **Transformers** use mutual inductance to change AC voltages and currents.

13.5.1 Challenge Problems

1. **Beginner:** An electron moves at 10^6 m/s perpendicular to a 0.05 T magnetic field. What is the radius of its circular path?
2. **Intermediate:** A rectangular conducting loop of width w is pulled at a constant velocity v out of a region of uniform magnetic field B . Find the induced EMF and the force required to pull it (assuming the loop has resistance R).
3. **Advanced (Olympiad):** A long, straight wire carrying current I lies in the plane of a rectangular loop, parallel to its long side. The loop has dimensions $a \times b$ and its nearest edge is at distance c from the wire. Calculate the mutual inductance M between the wire and the loop.

Chapter 14

Modern Physics & Cosmology

At the dawn of the 20th century, two major revolutions shattered classical physics: Quantum Mechanics (dealing with the very small) and Relativity (dealing with the very fast and very massive).

14.1 The Photoelectric Effect

When light shines on a metal surface, electrons can be ejected. Classical wave theory predicted that brighter light should increase the energy of the ejected electrons and that very dim light would take time to build up enough energy. Experiments proved both predictions wrong.

Einstein explained this by proposing that light itself is quantized into packets of energy called **photons**.

$$E = hf = \frac{hc}{\lambda}$$

where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant.

Key Formula: Photoelectric Equation

The maximum kinetic energy $K_{\max}$ of an ejected electron is:

$$K_{\max} = hf - \Phi$$

where Φ is the **work function** (the minimum energy required to tear an electron from the metal surface). If $hf < \Phi$, no electrons are ejected, regardless of the light intensity.

14.2 Wave-Particle Duality and the Bohr Model

Compton Scattering: Arthur Compton showed that X-ray photons scattering off electrons behave like colliding billiard balls, possessing momentum: $p = \frac{E}{c} = \frac{h}{\lambda}$.

De Broglie Wavelength: Louis de Broglie hypothesized the converse—if waves act like particles, particles should act like waves! Every particle with momentum p has a wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

The Bohr Model of the Atom: Niels Bohr used the wave nature of electrons to explain why atoms are stable and emit discrete spectral lines. He postulated that the angular momentum of an electron in a hydrogen atom is quantized:

$$L = mvr = n \frac{h}{2\pi}$$

This leads to quantized energy levels for the Hydrogen atom:

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

When an electron drops from a higher state n_i to a lower state n_f , it emits a photon with energy $E = E_{n_i} - E_{n_f}$.

14.3 Special Relativity

Einstein's Special Theory of Relativity rests on two postulates:

1. The laws of physics are the same in all inertial reference frames.
2. The speed of light in a vacuum (c) is constant for all observers, regardless of the motion of the light source or observer.

These simple postulates have profound consequences when velocities approach c . Let v be the relative velocity between two frames, and define the Lorentz factor:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \geq 1$$

Time Dilation: Moving clocks tick slower. If Δt_0 is the proper time measured by an observer at rest relative to the clock, an outside observer measures:

$$\Delta t = \gamma \Delta t_0$$

Length Contraction: Moving objects are shortened along their direction of motion. If L_0 is the proper length measured at rest, the outside observer measures:

$$L = \frac{L_0}{\gamma}$$

Mass-Energy Equivalence:

$$E = \gamma mc^2$$

The rest energy is $E_0 = mc^2$. The kinetic energy is $K = E - E_0 = (\gamma - 1)mc^2$.

14.4 Cosmology

Cosmology is the study of the universe as a whole.

The Big Bang Theory: The universe began as a hot, dense singularity approximately 13.8 billion years ago and has been expanding ever since.

Hubble's Law: Edwin Hubble discovered that distant galaxies are moving away from us, and their recession velocity v is proportional to their distance d :

$$v = H_0 d$$

where $H_0 \approx 70 \text{ km/s/Mpc}$ is the Hubble constant. This proves the expansion of the universe.

Cosmic Microwave Background (CMB): The afterglow of the Big Bang. About 380,000 years after the Big Bang, the universe cooled enough for electrons to bind to protons (recombination). The light emitted at that moment has been stretched by cosmic expansion into microwave radiation, bathing the entire sky uniformly at a temperature of $T \approx 2.7 \text{ K}$.

Chapter Summary

- **Photons** are quanta of light with energy $E = hf$ and momentum $p = \frac{h}{\lambda}$.
- **De Broglie's relation** ($\lambda = \frac{h}{p}$) shows that matter exhibits wave properties.

- **Special Relativity** requires that c is constant, leading to **time dilation** ($\Delta t = \gamma \Delta t_0$) and **length contraction** ($L = \frac{L_0}{\gamma}$).
- **Hubble's Law** ($v = H_0 d$) and the **CMB** are key observational pillars of the Big Bang model.

14.4.1 Challenge Problems

1. **Beginner:** What is the de Broglie wavelength of an electron moving at 10^7 m/s? ($m_e = 9.11 \times 10^{-31}$ kg).
2. **Intermediate:** A spaceship travels at $v = 0.8c$ relative to Earth. The astronaut measures the trip to a distant star to take 3 years on her clock. How far away is the star according to Earth observers?
3. **Advanced (Olympiad):** In a Compton scattering experiment, a photon of wavelength λ scatters off a stationary electron by an angle θ . Use conservation of relativistic energy and momentum to derive the Compton shift formula:

$$\lambda' - \lambda = \left(\frac{h}{m_e c} \right) (1 - \cos \theta).$$

Chapter 15

Appendices

15.1 Appendix A: Master Formula Sheet

Vectors & Kinematics

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$$

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y)\hat{i} - (A_x B_z - A_z B_x)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \mathbf{a} = \frac{d\mathbf{v}}{dt}$$

Mechanics

$$\sum \mathbf{F} = m\mathbf{a} = \frac{d\mathbf{p}}{dt}$$

$$f_s \leq \mu_s N, \quad f_k = \mu_k N$$

$$\mathbf{r}_{\text{cm}} = \frac{1}{M} \sum m_i \mathbf{r}_i$$

$$\mathbf{J} = \int \mathbf{r} \times d\mathbf{p} = \Delta \mathbf{L}$$

$$\sum \tau = I\alpha = \frac{dL}{dt}$$

$$I = \sum m_i r_i^2, \quad I = I_{\text{cm}} + Md^2$$

Work, Energy, Gravitation

$$W = \int \mathbf{F} \cdot d\mathbf{r}, \quad W_{\text{net}} = \Delta K$$

$$\Delta U = -W_c, \quad E = K + U = \text{constant (if only conservative forces)}$$

$$\mathbf{F}_g = -G \frac{m_1 m_2}{r^2}, \quad U_g = -\frac{G m_1 m_2}{r}$$

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}, \quad v_{\text{orbit}} = \sqrt{\frac{GM}{r}}$$

$$T^2 = \frac{4\pi^2}{GM} r^3$$

Oscillations & Waves

$$x(t) = A \cos(\omega t + \varphi), \quad \omega = 2\pi f = \frac{2\pi}{T}$$

$$\omega = \sqrt{\frac{k}{m}} \quad (\text{spring}), \quad \omega = \sqrt{\frac{g}{L}} \quad (\text{pendulum})$$

$$v = f\lambda = \frac{\omega}{k}$$

$$f_o = f_s \frac{v \pm v_o}{v \mp v_s}$$

Thermodynamics

$$\Delta L = \alpha L_0 \Delta T, \quad Q = mc\Delta T, \quad Q = mL$$

$$PV = nRT = Nk_B T$$

$$K_{\text{avg}} = \frac{3}{2} k_B T$$

$$\Delta E_{\text{int}} = Q - W$$

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_H}$$

Optics

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$d \sin \theta = m\lambda \quad (\text{Double slit bright fringes})$$

Electricity & Magnetism

$$F_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$V = IR, \quad P = IV$$

$$\mathbf{F}_B = q(\mathbf{v} \times \mathbf{B}), \quad \mathbf{F}_B = I(\mathbf{L} \times \mathbf{B})$$

$$\oint \mathbf{B} \cdot d\mathbf{s} = \mu_0 I_{\text{enclosed}}$$

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Modern Physics

$$E = hf = \frac{hc}{\lambda}$$

$$K_{\text{max}} = hf - \Phi$$

$$\lambda = \frac{h}{p}$$

$$\Delta t = \gamma \Delta t_0, \quad L = \frac{L_0}{\gamma}$$

$$E = \gamma mc^2$$

15.2 Appendix B: Bibliography and Further Reading

To excel in the Bangladesh Physics Olympiad, students are encouraged to consult the following classic textbooks and resources:

Core Mechanics and General Physics:

- **Fundamentals of Physics** by David Halliday, Robert Resnick, and Jearl Walker. (Excellent for building a strong foundation in all topics).
- **University Physics with Modern Physics** by Hugh D. Young and Roger A. Freedman. (Comprehensive standard textbook with great examples).
- **Introduction to Classical Mechanics: With Problems and Solutions** by David Morin. (Highly recommended for advanced problem-solving techniques and Olympiad-level mechanics).

Advanced Topics (Category C level and beyond):

- **Electricity and Magnetism** by Edward M. Purcell and David J. Morin. (The gold standard for E&M at the Olympiad level).
- **An Introduction to Mechanics** by Daniel Kleppner and Robert Kolenkow. (A rigorous, mathematically demanding mechanics text).
- **Concepts of Physics (Vol 1 & 2)** by H.C. Verma. (A staple for conceptual clarity and challenging problems in the Indian subcontinent).

Olympiad Specific Materials:

- **200 Puzzling Physics Problems** by P. Gnädig, G. Honyek, and K. Riley. (A collection of brilliant problems that test deep physical intuition).
- Past papers of the Bangladesh Physics Olympiad (BdPhO) and the International Physics Olympiad (IPhO).

15.3 Appendix C: Index

Because this document is generated dynamically, a formal page-numbered index is not included in this digital format. However, the following key terms represent the core concepts covered in this book. Students should ensure they are deeply familiar with each:

- **A:** Acceleration, Adiabatic, Ampere's Law, Angular Momentum, Archimedes' Principle.
- **B:** Beats, Bernoulli's Equation, Big Bang, Biot-Savart Law, Bohr Model, Bulk Modulus.
- **C:** Calorimetry, Capacitance, Carnot Engine, Center of Mass, Collisions, Compton Scattering, Continuity Equation, Coulomb's Law, Cross Product.
- **D:** De Broglie Wavelength, Density, Dielectrics, Diffraction, Dispersion, Doppler Effect, Dot Product, Drag Force.
- **E:** Efficiency, Elastic Collision, Electric Field, Electric Potential, Electromagnetic Induction, Entropy, Equilibrium, Escape Velocity.
- **F:** Faraday's Law, First Law of Thermodynamics, Focal Length, Free-Body Diagram, Friction, Fundamental Frequency.
- **G:** Gauss's Law, Gravitational Field, Gravitational Potential.
- **H:** Heat Capacity, Heat Engine, Hooke's Law, Hubble's Law.
- **I:** Ideal Gas Law, Impulse, Index of Refraction, Inductance, Inelastic Collision, Inertia, Interference, Isothermal.
- **K:** Kepler's Laws, Kinematics, Kinetic Energy, Kinetic Friction, Kinetic Theory of Gases, Kirchhoff's Laws.
- **L:** Latent Heat, Lens Maker's Equation, Lenz's Law, Lorentz Factor, Lorentz Force.
- **M:** Magnetic Field, Magnetic Flux, Magnification, Moment of Inertia, Momentum, Mutual Inductance.
- **N:** Newton's Laws of Motion, Nodes.
- **O:** Ohm's Law, Optics, Oscillations.
- **P:** Parallel Axis Theorem, Pascal's Principle, Photoelectric Effect, Phase Velocity, Poiseuille's Law, Potential Energy, Power, Pressure, Projectile Motion.
- **R:** Refraction, Resistance, Resonance, Rest Energy, Rotational Motion.
- **S:** Scalar Product, Second Law of Thermodynamics, Self-Inductance, Shell Theorem, Simple Harmonic Motion (SHM), Snell's Law, Special Relativity, Static Friction, Superposition, Surface Tension.
- **T:** Temperature, Terminal Velocity, Thermal Expansion, Thermodynamics, Torque, Total Internal Reflection, Transformers.
- **V:** Vector Addition, Velocity, Viscosity.
- **W:** Wave Equation, Wavelength, Work, Work-Energy Theorem, Work Function.
- **Y:** Young's Double-Slit Experiment.